

Parking Aid — Video



Special Tool(s)	
 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent

Principles of Operation

NOTE: The tailgate must be closed for correct operation of the rear video camera system.

Rear Video Camera

The rear video camera system is active with the ignition in run and the transmission selector in the REVERSE (R). The rear video camera receives commands and sends status to the Body Control Module (BCM) over a dedicated Local Interconnect Network (LIN) . For vehicles with a 203.2 mm (8 in) touchscreen, the rear video camera sends a video signal to the APIM on a pair of shielded video circuits. The APIM then displays the rear video camera image on the FDIM . For vehicles without a 203.2 mm (8 in) touchscreen, the signal from the rear video camera is sent to the auto-dimming interior mirror on a pair of shielded video circuits, which displays the video image on the LH side of the auto-dimming interior mirror glass.

If the vehicle is equipped with a front video camera, the rear video camera sends the rear video signal to the camera switching module on a pair of shielded video circuits. The camera switching module determines which video image should be displayed based on vehicle operating conditions, and then sends the front or rear video camera signal to the APIM on a pair of shielded video circuits.

The rear video camera system has 4 video features (only the fixed guidelines are available on vehicles equipped with the base Instrument Panel Cluster (IPC)). The features are:

- Fixed guidelines
- Visual park aid alert
- Manual zoom
- Video camera delay

Vehicles equipped with a base IPC (2 rows, 14 character display), display only fixed guidelines in the auto-dimming interior mirror. The guidelines cannot be disabled.

Vehicles equipped with an optional **IPC** (4.2 LCD message center) include all the rear video camera system features. The features are enabled and disabled within the message center.

For vehicles with a 203.2 mm (8 in) touchscreen, the rear video camera features are controlled within the **FDIM** menu screen. When the driver changes any of the rear video camera system features, the **APIM** sends a message to the **BCM** via the Medium Speed Controller Area Network (MS-CAN) . The **BCM** sends the message to the rear video camera over the **LIN** . The **LIN** is a dedicated single wire communication line that sends and receives network messages from the **BCM** to the rear video camera.

The messages sent from the **BCM** to the rear video camera are:

- transmission selector (PRNDL) status
- camera configuration data
- display manual zoom request
- parking aid audible warning status
- parking aid sensor distance to object data

The messages sent from the rear video camera to the **BCM** are:

- camera status
- display zoom status
- camera part number data
- visual park aid alert status
- fixed guideline status

The rear video system has a delay feature and is available on vehicles with optional Instrument Panel Cluster (IPC) or a 203.2 mm (8 in) touchscreen. For vehicles equipped with a 203.2 mm (8 in) touchscreen, the image is displayed in the **FDIM** for 5 minutes or until the vehicle reaches 8 km/h (5 mph), when the vehicle is shifted out of REVERSE (R), or if any button on the **FDIM** is pressed. The image is not displayed if the vehicle is shifted into PARK (P). For vehicles without a 203.2 mm (8 in) touchscreen, the image is displayed in the auto-dimming interior mirror and remains on for a few seconds once the vehicle is shifted out of REVERSE (R).

The fixed guidelines feature displays guidelines on top of the video image to assist the driver with alignment of the vehicle. A dashed line on the displayed image represents the center of the vehicle and 3 color-coded lines (red, yellow, green) identify different zones between the rear of the vehicle and objects.

NOTE: *The color-coded lines cannot indicate accurate or consistent distances between objects located behind the vehicle and at the rear of the vehicle. This normal condition is due to variances in vehicle ride height, including, but not limited to, vehicle loading.*

The visual park aid alert feature utilizes the 4 parking aid sensor inputs from the Parking Aid Module (PAM) and the rear video camera image. When the transmission selector is in REVERSE (R) and an object is detected in the path of the parking aid sensor, a network message of the distance from the parking aid sensor to the object is sent from the **PAM** to the **BCM** over the **HS-CAN** . The **BCM** networks the parking aid sensor message to the rear video camera over the **LIN** . The visual alerts are red, yellow or green and appear as highlights on top of the displayed image to indicate in which zone the object is detected. The highlighted visual alerts are still operational if the parking aid system is disabled within the **IPC** and the visual alert feature is enabled.

The manual zoom feature is controlled by the rear video camera. The manual zoom feature has 3 zoom levels that are only active when the vehicle is in reverse. If the manual zoom feature is enabled and the vehicle is shifted out of reverse the manual zoom feature is disabled. The manual zoom feature must be re-enabled the next time the vehicle is shifted into reverse. For vehicles with a 203.2 mm (8 in) touchscreen, when the driver inputs to the **FDIM** request the video image to be zoomed in or out, the **APIM** sends a zoom request network message to the **BCM** . For vehicles without a 203.2 mm (8 in) touchscreen, when the message center requests the video image to be zoomed in or out, the **IPC** sends a zoom request network message to the **BCM** . The **BCM** sends the zoom request message to the rear video camera over the **LIN** . The video image sent by the rear video camera is then digitally zoomed in or out.

Front Video Camera — F-150 Raptor Only

The front video camera system is active with the ignition in run, the transmission selector in the drive position, the vehicle is in off road mode and the Electronic Locking Differential (ELD) or 4L (4X4 LOW) is activated.

The front video camera receives commands and sends status to the Body Control Module (BCM) over a dedicated Local Interconnect Network (LIN) . The front video camera sends a video signal to the camera switching module on a pair of shielded video circuits. The camera switching module determines which video image should be displayed based on vehicle operating conditions, and then sends the front or rear video camera signal to the **APIM** on a pair of shielded video circuits.

To help distinguish the front video display from the rear video display, there is an icon displayed in the upper left corner of the screen for the front camera. It is an image of a camera with the letter "F".

The front video camera system has 3 video features (only the fixed guidelines are available with the base Instrument Panel Cluster (IPC)). The features are:

- Fixed guidelines
- Visual park aid alert
- Manual zoom

The front video camera features are controlled within the **FDIM** menu screen. For each one of these system features the Audio Front Control Module (ACM) sends a network message requesting the system feature to the **BCM** via the Medium Speed Controller Area Network (MS-CAN) . The **BCM** sends the message to the front video camera over the **LIN** . The **LIN** is a dedicated single wire communication line that sends and receives network messages from the **BCM** to the front video camera.

The messages sent from the **BCM** to the front video camera are:

- transmission selector (PRNDL) status
- camera configuration data
- display manual zoom request
- 4L active
- Electronic Locking Differential (ELD) active

The fixed guidelines feature displays guidelines on top of the video image to assist the driver with alignment of the vehicle. A dashed line on the displayed image represents the center of the vehicle and 3 color-coded lines (red, yellow, green) identify different zones between the front of the vehicle and objects.

NOTE: *The color-coded lines cannot indicate accurate or consistent distances between objects located in front of the vehicle and at the front of the vehicle. This normal condition is due to variances in vehicle ride height, including, but not limited to, vehicle loading.*

The manual zoom feature is controlled by the front video camera. The manual zoom feature has 3 zoom levels that are only active when the vehicle is in a forward gear, and either the 4L or **ELD** are active. The manual zoom feature must be re-enabled the next time the vehicle is shifted into a forward gear. When the **FDIM** requests the video image to be zoomed in or out, the **ACM** sends a zoom request network message to the **BCM** . The **BCM** sends the zoom request message to the front video camera over the **LIN** . The video image is then digitally zoomed in or out.

Camera Switching Module

The rear video camera always sends a video signal to the camera switching module. The front video camera only sends a video signal to the camera switching module when the vehicle is in a forward gear and the vehicle is in 4L or **ELD** modes. When the camera switching module receives only a rear video signal, the rear video signal is sent to the **APIM** . When the camera switching module receives both front and rear signals, the front video signal is sent to the **APIM** . An icon of a camera with the letter "F" is displayed while the front video signal is active.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical and electrical damage.

Visual Inspection Chart

Mechanical	Electrical
• Tailgate and rear video camera • Grille and front video camera	• Body Control Module (BCM) fuse(s): ▪ 32 (15A) (vehicles without a 203.2 mm (8 in) touchscreen) ▪ 34 (10A) • Wiring, terminals and connectors • Rear video camera • Front video camera • Auto-dimming interior mirror (without a 203.2 mm (8 in) touchscreen) • Front Display Interface Module (FDIM) (with a 203.2 mm (8 in) touchscreen) • Accessory Protocol Interface Module (APIM) (with a 203.2 mm (8 in) touchscreen)

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

4. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

5. **NOTE:** *The Vehicle Communication Module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM .*

If the scan tool still does not communicate with the VCM :

- check the VCM connection to the vehicle.
- check the scan tool connection to the VCM .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

6. If the scan tool does not communicate with the vehicle:

- verify the ignition key is in the RUN position.
- verify the scan tool operation with a known good vehicle.
- refer to Section 418-00, The PCM Does Not Respond To The Scan Tool, to diagnose no response from the PCM.

7. Carry out the network test.

- If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
- If the network passes, retrieve and record the continuous memory DTCs.

8. Clear the continuous DTCs and carry out the self-test diagnostics for the BCM and Audio Front Control Module (ACM) (vehicles with navigation).

9. If the DTCs retrieved are related to the concern, go to DTC Charts. For all other DTCs, refer to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.

10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

DTC Charts

APIM DTC Chart

NOTE: *This module utilizes a 5-character DTC followed by a 2-character failure-type code. The failure-type code provides information about specific fault conditions such as opens, or shorts to ground. Continuous memory DTCs have an additional 2-character DTC status code suffix to assist in determining DTC history.*

DTC	Description	Action
C1001:01	Vision System Camera: General Electrical Failure	For F-150 Raptor equipped with front video camera, GO to Pinpoint Test I . For all other vehicles, GO to Pinpoint Test E .
All other DTCs	—	REFER to the Diagnostic Trouble Code (DTC) Charts in Section 415-00E or Section 415-00F.

Body Control Module (BCM) DTC Chart

NOTE: This module utilizes a 5-character DTC followed by a 2-character failure-type code. The failure-type code provides information about specific fault conditions such as opens, or shorts to ground. Continuous memory DTCs have an additional 2-character DTC status code suffix to assist in determining DTC history.

DTC	Description	Action
B115E:01	Camera Module: General Electrical Failure	DTC B115E:01 sets in the BCM when the rear video camera status indicates corrupted data with in the rear video camera. CLEAR the DTCs. REPEAT the self-test. If DTC B115E:01 is retrieved again, INSTALL a new rear video camera. REFER to Video Camera — Rear in this section. TEST the system for normal operation.
B115E:08	Camera Module: Bus Signal/Message Failure	GO to Pinpoint Test E . For F-150 Raptor equipped with front video camera, GO to Pinpoint Test I .
B115E:49	Camera Module: Internal Electronic Failure	CLEAR the DTCs. REPEAT the self-test. If DTC B115E:49 is retrieved again, INSTALL a new rear video camera. REFER to Video Camera — Rear in this section. TEST the system for normal operation.
B115E:55	Camera Module: Not Configured	DTC B115E:55 sets in the BCM when the rear video camera is not configured. The rear video camera needs to be configured through the BCM. CARRY OUT the rear video camera configuration using the scan tool. TEST the system for normal operation.
B1331:02	Compass/Mirror Module: General Signal Failure	GO to Pinpoint Test K .
B1331:08	Compass/Mirror Module: Bus Signal/Message Failure	GO to Pinpoint Test K .
B1331:49	Compass/Mirror Module: Internal Electronic Failure	Sets in continuous memory when the BCM detects an internal fault in the auto-dimming interior mirror. CLEAR the DTCs and test the system for normal operation. If DTC B1331:49 returns, INSTALL a new auto-dimming interior mirror. Section 501-09.
All other DTCs	—	REFER to the DTC Charts in Section 419-10.

Video Symptom Chart

Symptom Chart

Condition	Possible Causes	Action
The rear video is inoperative	- Wiring, terminals or connectors - Rear video camera - Front video camera - APIM (with a 203.2 mm (8 in) touchscreen) - Auto-dimming interior mirror (without a 203.2 mm (8 in) touchscreen) - Video camera configuration issue - Body Control Module (BCM)	- For F-150 Raptor equipped with front video camera, GO to Pinpoint Test I. - For all other vehicles, GO to Pinpoint Test E.
The front video is inoperative	- Wiring, terminals or connectors - Front video camera - Rear video camera - Video camera configuration issue - APIM - Body Control Module (BCM)	- If the rear camera needs to be configured or DTC B115E:55 is present. Diagnose this concern first. The front video camera is inoperative if configuration is required for the rear camera. GO to Pinpoint Test I.
Black screen or no image	- Fuse - Front video camera - Rear video camera - Wiring, terminals or connectors - Auto-dimming interior mirror (without a 203.2 mm (8 in) touchscreen) - APIM (with a 203.2 mm (8 in) touchscreen)	- For F-150 Raptor equipped with front video camera, GO to Pinpoint Test I. - For all other vehicles, GO to Pinpoint Test E.

• Rear video displayed when the front video should be displayed	• Fuse • Front video camera • Wiring, terminals or connectors	• GO to Pinpoint Test I.
• Blue Screen (mirror only)	• Fuse • Rear video camera • Wiring, terminals or connectors	• GO to Pinpoint Test E.
• The Front Display Interface Module (FDIM) displays PLEASE CHECK SURROUNDINGS FOR SAFETY and a black screen/no image	• Fuse • Wiring, terminals or connectors • Front video camera • Rear video camera • BCM	• NOTE: <i>It may take up to 30 seconds before the message is displayed on the FDIM .</i> • For F-150 Raptor equipped with front video camera, GO to Pinpoint Test I. • For all other vehicles, GO to Pinpoint Test E.
• Front camera static/flashing/blinking/flickering/scrolling or rolling image/screen/scrolling horizontal lines	• Front video camera • Wiring, terminals or connectors	• GO to Pinpoint Test J.
• Front camera poor image quality (foggy/cloudy/fuzzy/blurry/hazy image or moisture/water in lens or black spots)	• Front video camera • Wiring, terminals or connectors	• GO to Pinpoint Test J.
• Rear camera static/flashing/blinking/flickering/scrolling or rolling image/screen/scrolling horizontal lines	• Rear video camera • Wiring, terminals or connectors	• For F-150 Raptor equipped with front video camera, GO to Pinpoint Test J. • For all other vehicles, GO to Pinpoint Test F.
• Rear camera poor image quality (foggy/cloudy/fuzzy/blurry/hazy image or moisture/water in lens or black spots)	• Rear video camera • Wiring, terminals or connectors	• For F-150 Raptor equipped with front video camera, GO to Pinpoint Test J. • For all other vehicles, GO to Pinpoint Test F.
• Manual zoom function is inoperative	• Rear video camera not configured after	• NOTE: <i>A LIN circuit fault may prevent the</i>

	replacement (DTC B115E:55 may be present) • Rear video camera • Manual zoom feature is disabled	<i>front or rear camera from being configured properly. VERIFY the integrity of the wiring, terminals and connectors.</i> • CARRY OUT the rear video camera configuration using the scan tool. • For F-150 Raptor equipped with front video camera, GO to Pinpoint Test I. • For all other vehicles, GO to Pinpoint Test E.
• Visual park aid alert is inoperative	• Rear video camera not configured after replacement (DTC B115E:55 may be present) • Rear video camera • Wiring, terminals or connectors • PAM • BCM	• NOTE: A LIN circuit fault may prevent the front or rear camera from being configured properly. VERIFY the integrity of the wiring, terminals and connectors. • CARRY OUT the rear video camera configuration using the scan tool. • GO to Pinpoint Test H.
• Fixed guidelines are not displayed on the video image	• Rear video camera not configured after replacement (DTC B115E:55 may be present)	• NOTE: A LIN circuit fault may prevent the front or rear camera from being configured properly. VERIFY the integrity of the wiring, terminals and connectors. • CARRY OUT the rear video camera configuration using the scan tool.

	• Rear video camera • Fixed guidelines feature is disabled	• For F-150 Raptor equipped with front video camera, GO to Pinpoint Test I. • For all other vehicles, GO to Pinpoint Test E.
• Rear video camera delay feature is inoperative	• Rear video camera delay feature is disabled	• ENABLE the rear video camera delay feature within the FDIM (with a 203.2 mm (8 in) touchscreen) or message center (without a 203.2 mm (8 in) touchscreen and optional cluster).
• The rear video display is always on (vehicles without a 203.2 mm (8 in) touchscreen)	• Wiring, terminals or connectors • Auto-dimming interior mirror	• GO to Pinpoint Test G.
• The rear video display is always on (vehicles with a 203.2 mm (8 in) touchscreen)	• Reverse lamp input always on	• For F-150 Raptor equipped with front video camera, GO to Pinpoint Test I. • For all other vehicles, GO to Pinpoint Test E.
• All other image concerns	• Rear video camera • Front video camera	• For F-150 Raptor equipped with front video camera, GO to Pinpoint Test I. • For all other vehicles, GO to Pinpoint Test E.

• Front camera displayed when in reverse	• Front or rear video camera not configured after replacement (DTC B115E:55 may be present)	• NOTE: A LIN circuit fault may prevent the front or rear camera from being configured properly. <i>VERIFY</i> the integrity of the wiring, terminals and connectors. • CARRY OUT the rear video camera configuration using the scan tool.
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Pinpoint Tests

Pinpoint Test E: The Rear Video Does Not Operate Correctly

Normal Operation

When the ignition switch is in the RUN position, the rear video camera receives voltage. For vehicles without a 203.2 mm (8 in) touchscreen, with the transmission selector in REVERSE (R), a reverse signal message from the Body Control Module (BCM) is sent over a dedicated Local Interconnect Network (LIN) to the auto-dimming interior mirror, enabling the system. For vehicles with a 203.2 mm (8 in) touchscreen, with the transmission selector in REVERSE (R), the BCM sends a reverse signal message to the APIM over the Medium Speed Controller Area Network (MS-CAN) , enabling the system. The video signal from the rear video camera displays an image on the left portion of the auto-dimming interior mirror for vehicles without a 203.2 mm (8 in) touchscreen, or displays an image on the Front Display Interface Module (FDIM) for vehicles with a 203.2 mm (8 in) touchscreen.

- DTC B115E:08 (Camera Module: Bus Signal/Message Failure) — an on-demand DTC that sets in the BCM when there is a ~~LIN~~ communication fault from the rear video camera.
- DTC C1001:01 (Vision System Camera: General Electrical Failure) — a continuous DTC that sets in the APIM when no video signal is detected from the rear video camera when the vehicle is in reverse (vehicles with a 203.2 mm (8 in) touchscreen only).

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Rear video camera
- Camera switching module (with front camera)
- Auto-dimming interior mirror (without a 203.2 mm (8 in) touchscreen)
- Accessory Protocol Interface Module (APIM) (with a 203.2 mm (8 in) touchscreen)
- BCM

PINPOINT TEST E : THE REAR VIDEO DOES NOT OPERATE CORRECTLY

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: <i>Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.</i>
E1 CHECK AND CLEAN THE REAR VIDEO CAMERA LENS
• Clean and make sure the rear video camera lens is clear of any debris. • Test the rear video camera system and verify the rear video camera image.

Is the rear video camera concern still present?

Yes	GO to E2 .
No	The system is operating correctly at this time. The concern was caused by a dirty rear video camera lens.

E2 CHECK THE REVERSING LAMPS OPERATION

- Ignition OFF.
- Apply the parking brake.
- Ignition ON.
- Select REVERSE.
- Check the operation of the reversing lamps.

Do the reversing lamps operate correctly?

Yes	GO to E3 .
No	REFER to Section 417-01.

E3 CHECK THE DTCS FROM THE BCM SELF-TEST

- Check the ~~BCM~~ DTCs from the self-test.

Are any ~~BCM~~ DTCs retrieved?

Yes	For DTC B115E:08, GO to E5 . For all other DTCs, REFER to DTC Charts in this section.
No	For an inoperative rear video camera feature concern, GO to E4 . Otherwise, GO to E5 .

E4 VERIFY AND ENABLE THE INOPERATIVE REAR VIDEO CAMERA FEATURE

- Verify and enable the inoperative rear video camera feature. Refer to the Owner's Literature.

Does the rear video camera feature operate correctly?

Yes	The rear video camera system is operating correctly at this time. The rear video camera feature concern was caused by the feature being disabled. INSTRUCT the customer on how to enable the rear video camera features, REFER to the Owner's Literature.
No	GO to E5 .

E5 VERIFY VEHICLE EQUIPMENT — 203.2 MM (8 IN) TOUCHSCREEN

- Check the vehicle for a 203.2 mm (8 in) touchscreen.

Is the vehicle equipped with a 203.2 mm (8 in) touchscreen?

Yes	GO to E10 .
No	GO to E6 .

E6 CHECK THE AUTO-DIMMING INTERIOR MIRROR

- Observe the auto-dimming interior mirror.
- Select PARK.
- Select REVERSE.

Does the auto-dimming interior mirror momentarily display a blue screen then turn off?

Yes	The auto-dimming interior mirror is functional. There is no signal from the rear video camera. GO to E11 .
No	GO to E7 .

E7 CHECK THE COMPASS FOR CORRECT OPERATION

- **NOTE:** *The auto-dimming interior mirror uses the LIN for compass messaging along with messaging for the rear video camera operation.*

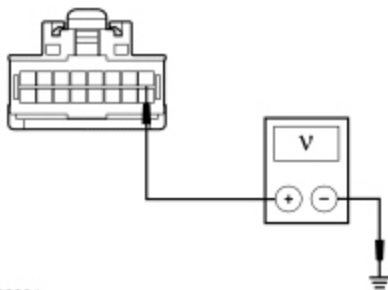
Verify the compass operates correctly.

Does the compass operate correctly?

Yes	GO to E8 .
No	REFER to Section 413-01 to diagnose the compass.

E8 CHECK THE AUTO-DIMMING INTERIOR MIRROR VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Auto-Dimming Interior Mirror [C9039](#).
- Ignition ON.
- Measure the voltage between the auto-dimming interior mirror [C9039](#) Pin 1, circuit CBP32 (GN/VT), harness side and ground.



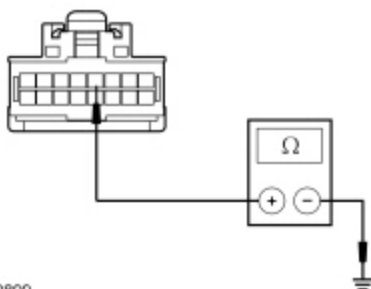
N0068991

Is the voltage greater than 10 volts?

Yes	GO to E9 .
No	VERIFY the BCM fuse 32 (15A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. TEST the system for normal operation.

E9 CHECK THE AUTO-DIMMING INTERIOR MIRROR GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the auto-dimming interior mirror [C9039](#) Pin 4, circuit GD133 (BK), harness side and ground.



N0120899

Is the resistance less than 5 ohms?

Yes	INSTALL a new auto-dimming interior mirror. REFER to Section 501-09. CONNECT the negative battery cable. TEST the system for normal operation.
No	REPAIR the circuit. CONNECT the negative battery cable. TEST the system for normal operation.

E10 CHECK THE FDIM DISPLAY

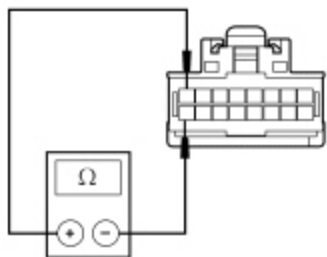
- Observe the FDIM while placing the vehicle in reverse.

While selecting reverse and waiting up to 30 seconds, does the FDIM screen switch to rear video camera mode and display "Please Contact Your Dealership"?

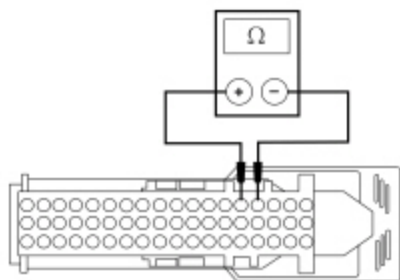
Yes	The FDIM display is functional. There is no signal from the rear video camera. GO to E11 .
No	RETRIEVE and FOLLOW audio system DTCs. REFER to Section 415-00E or Section 415-00F.

E11 CHECK THE VIDEO SIGNAL CIRCUITS INTEGRITY

- Ignition OFF.
- Disconnect: Auto-Dimming Interior Mirror (Without Navigation) [C9039](#).
- Disconnect: APIM (With a 203.2 mm (8 in) touchscreen) [C2383](#).
- For vehicles without a 203.2 mm (8 in) touchscreen, measure the resistance between the auto-dimming interior mirror [C9039](#) Pin 8, circuit VMP19 (WH/GN), harness side and the auto-dimming interior mirror [C9039](#) Pin 16, circuit RMP19 (BN/VT), harness side.



- N0069002
- For vehicles with a 203.2 mm (8 in) touchscreen, measure the resistance between the APIM [C2383](#) Pin 14, circuit VMP19 (WH/GN), harness side and the APIM [C2383](#) Pin 15, circuit RMP19 (BN/VT), harness side.



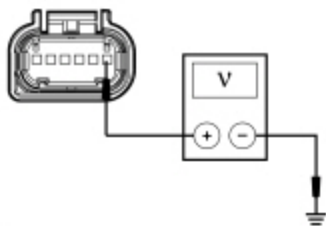
• N0149434

Is the resistance between 74 and 106 ohms?

Yes	GO to E12 .
No	GO to E14 .

E12 CHECK THE REAR VIDEO CAMERA VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Disconnect: Rear Video Camera [C4357](#).
- Ignition ON.
- Measure the voltage between the rear video camera [C4357](#) Pin 1, circuit CBP34 (VT/BN), harness side and ground.



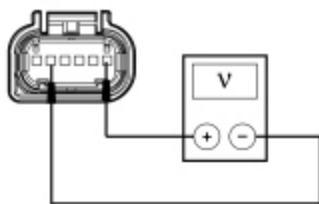
N0120901

Is the voltage greater than 10 volts?

Yes	GO to E13 .
No	VERIFY the BCM fuse 34 (10A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. TEST the system for normal operation.

E13 CHECK THE REAR VIDEO CAMERA GROUND CIRCUIT FOR AN OPEN

- Measure the voltage between the rear video camera [C4357](#) Pin 1, circuit CBP34 (VT/BN), harness side and the rear video camera [C4357](#) Pin 5, circuit GD120 (BK/GN), harness side and ground.



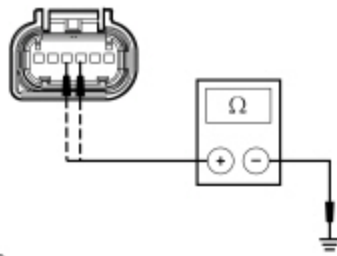
N0120902

Is the voltage greater than 10 volts?

Yes	GO to E14 .
No	REPAIR circuit GD120 (BK/GN) for an open. TEST the system for normal operation.

E14 CHECK THE REAR VIDEO CAMERA SIGNAL CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: Rear Video Camera [C4357](#).
- Measure the resistance between the rear video camera [C4357](#) Pin 4, circuit VMP19 (WH/GN), harness side and ground; and between the rear video camera [C4357](#) Pin 3, circuit RMP19 (BN/VT), harness side and ground.



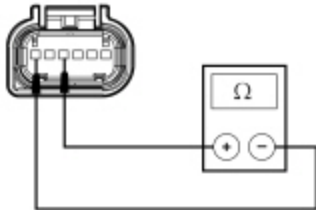
N0121053

Are the resistances greater than 10,000 ohms?

Yes	GO to E15 .
No	REPAIR the circuit in question. TEST the system for normal operation.

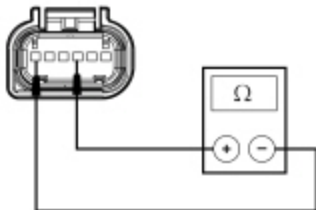
E15 CHECK THE REAR VIDEO CAMERA VIDEO SIGNAL CIRCUITS FOR A SHORT TO VIDEO SHIELD

- Measure the resistance between the rear video camera [C4357](#) Pin 4, circuit VMP19 (WH/GN), harness side and the rear video camera [C4357](#) Pin 6, circuit DMP19 (BK), harness side.



N0120904

- Measure the resistance between the rear video camera [C4357](#) Pin 3, circuit RMP19 (BN/VT), harness side and the rear video camera [C4357](#) Pin 6, circuit DMP19 (BK), harness side.



N0120905

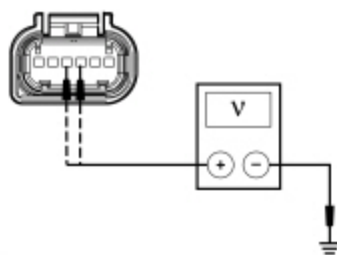
Are the resistances greater than 10,000 ohms?

Yes	GO to E16 .
No	REPAIR the circuit in question. TEST the system for normal operation.

E16 CHECK THE REAR VIDEO CAMERA VIDEO SIGNAL CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition ON.

- Measure the voltage between the rear video camera [C4357](#) Pin 4, circuit VMP19 (WH/GN), harness side and ground; and between the rear video camera [C4357](#) Pin 3, circuit RMP19 (BN/VT), harness side and ground.



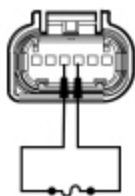
N0120909

Is any voltage present?

Yes	REPAIR the circuit in question. TEST the system for normal operation.
No	GO to E17 .

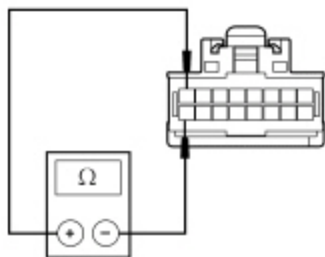
E17 CHECK VIDEO SIGNAL CIRCUITS FOR AN OPEN

- Ignition OFF.
- Connect a fused jumper wire between the rear video camera [C4357](#) Pin 4, circuit VMP19 (WH/GN), harness side and the rear video camera [C4357](#) Pin 3, circuit RMP19 (BN/VT), harness side.



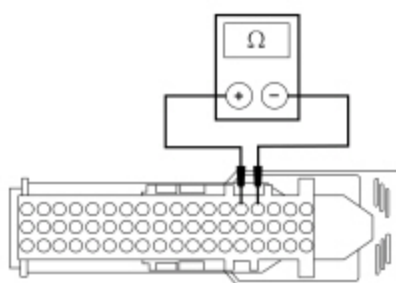
N0120906

- For vehicles without a 203.2 mm (8 in) touchscreen, measure the resistance between the auto-dimming interior mirror [C9039](#) Pin 8, circuit VMP19 (WH/GN), harness side and the auto-dimming interior mirror [C9039](#) Pin 16, circuit RMP19 (BN/VT), harness side.



N0069002

- For vehicles with a 203.2 mm (8 in) touchscreen, measure the resistance between the APIM [C2383](#) Pin 14, circuit VMP19 (WH/GN), harness side and the APIM [C2383](#) Pin 15, circuit RMP19 (BN/VT), harness side.



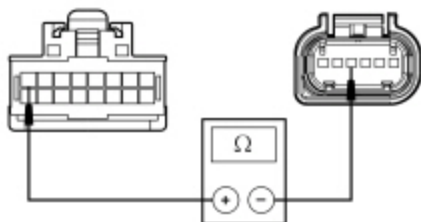
N0149434

Is the resistance less than 5 ohms?

Yes	REMOVE the jumper wire. GO to E19 .
No	REMOVE the jumper wire. GO to E18 .

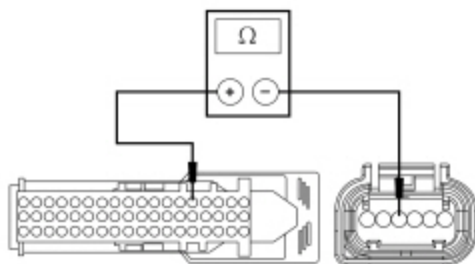
E18 CHECK VIDEO (+) SIGNAL CIRCUIT FOR AN OPEN

- For vehicles without a 203.2 mm (8 in) touchscreen, measure the resistance between the auto-dimming interior mirror [C9039](#) Pin 8, circuit VMP19 (WH/GN), harness side and the rear video camera [C4357](#) Pin 4, circuit VMP19 (WH/GN), harness side.



N0120907

- For vehicles with a 203.2 mm (8 in) touchscreen, measure the resistance between the APIM [C2383](#) Pin 14, circuit VMP19 (WH/GN), harness side and the video camera [C4357](#) Pin 4, circuit VMP19 (WH/GN), harness side.



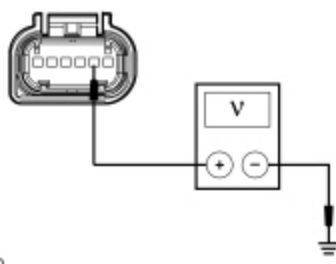
N0149435

Is the resistance less than 5 ohms?

Yes	REPAIR circuit RMP19 (BN/VT). TEST the system for normal operation.
No	REPAIR circuit VMP19 (WH/GN). TEST the system for normal operation.

E19 CHECK THE LIN CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Rear Video Camera [C4357](#).
- Disconnect: BCM [C2280C](#) .
- Ignition ON.
- Measure the voltage between the rear video camera [C4357](#) Pin 2, circuit VDN03 (GY/VT), harness side and ground.



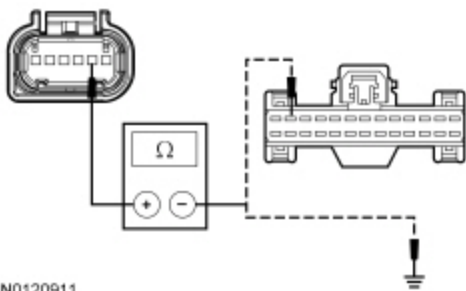
N0120910

Is any voltage present?

Yes	REPAIR the circuit. TEST the system for normal operation.
No	GO to E20 .

E20 CHECK THE LIN CIRCUIT FOR AN OPEN OR SHORT TO GROUND

- Ignition OFF.
- Measure the resistance between the rear video camera [C4357](#) Pin 2, circuit VDN03 (GY/VT), harness side and the BCM [C2280C](#) Pin 12, circuit VDN03 (GY/VT), harness side; and between the rear video camera [C4357](#) Pin 2, circuit VDN03 (GY/VT), harness side and ground.



N0120911

Is resistance less than 5 ohms between the rear video camera and BCM ; and greater than 10,000 ohms between the rear video camera and ground?

Yes	GO to E21 .
No	REPAIR the circuit. TEST the system for normal operation.

E21 CHECK FOR CORRECT REAR VIDEO CAMERA OPERATION

- Connect: All Disconnected Connectors .

- Disconnect the rear video camera connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the rear video camera connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new rear video camera. REFER to Video Camera — Rear in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test F: Poor Image Quality — Rear Video Camera

Normal Operation

A pair of shielded wires carries the video signal from the rear video camera to the Front Display Interface Module (FDIM) for vehicles with navigation, and the auto-dimming interior mirror for vehicles without navigation. The rear video system provides a clear image of objects to the rear of the vehicle.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Rear video camera

PINPOINT TEST F : POOR IMAGE QUALITY

NOTE: Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.

F1 CHECK AND CLEAN THE REAR VIDEO CAMERA LENS

- Clean and make sure the rear video camera lens is clear of any debris.
- Test the rear video camera system and verify the rear video camera image.

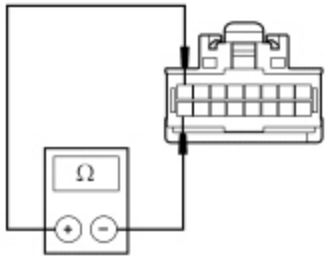
Is the rear video camera concern still present?

Yes	GO to F2 .
No	The system is operating correctly at this time. The concern was caused by a dirty rear video camera lens.

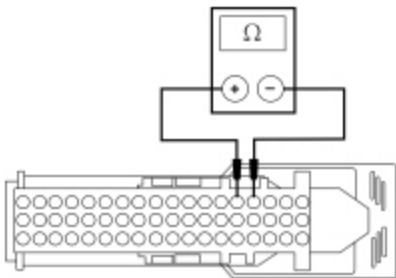
F2 CHECK THE REAR VIDEO SIGNAL CIRCUITS INTEGRITY

- Ignition OFF.
- Disconnect: Auto-Dimming Interior Mirror (Without a 203.2 mm (8 in) touchscreen) [C9039](#).

- Disconnect: APIM (With a 203.2 mm (8 in) touchscreen) [C2123](#).
- For vehicles without a 203.2 mm (8 in) touchscreen, measure the resistance between the auto-dimming interior mirror [C9039](#) Pin 8, circuit VMP19 (WH/GN), harness side and the auto-dimming interior mirror [C9039](#) Pin 16, circuit RMP19 (BN/VT), harness side.



- N0069002
- For vehicles with a 203.2 mm (8 in) touchscreen, measure the resistance between the APIM [C2383](#) Pin 14, circuit VMP19 (WH/GN), harness side and the APIM [C2383](#) Pin 15, circuit RMP19 (BN/VT), harness side.



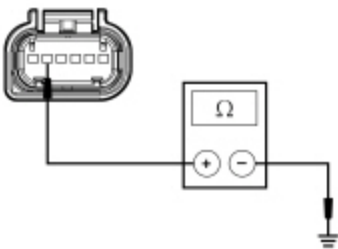
• N0149434

Is the resistance between 74 and 106 ohms?

Yes	GO to F3 .
No	GO to F4 .

F3 CHECK THE REAR VIDEO CAMERA GROUND CIRCUIT FOR AN OPEN

- Disconnect: Negative Battery Cable .
- Disconnect: Rear Video Camera [C4357](#).
- Measure the resistance between the rear video camera [C4357](#) Pin 5, circuit GD120 (BK/GN), harness side and ground.



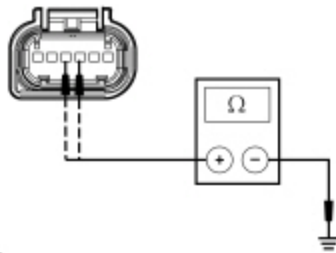
• N0120912

Is the resistance less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to F4 .
No	REPAIR the circuit. CONNECT the negative battery cable. TEST the system for normal operation.

F4 CHECK THE REAR VIDEO CAMERA SIGNAL CIRCUITS FOR A SHORT TO GROUND

- Disconnect: Rear Video Camera [C4357](#).
- Measure the resistance between the rear video camera [C4357](#) Pin 4, circuit VMP19 (WH/GN), harness side and ground; and between the rear video camera [C4357](#) Pin 3, circuit RMP19 (BN/VT), harness side and ground.



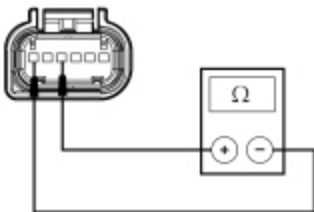
N0121053

Are the resistances greater than 10,000 ohms?

Yes	GO to F5 .
No	REPAIR the circuit in question. TEST the system for normal operation.

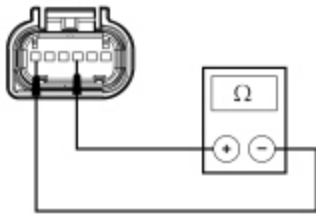
F5 CHECK THE REAR VIDEO CAMERA VIDEO SIGNAL CIRCUITS FOR A SHORT TO VIDEO SHIELD

- Measure the resistance between the rear video camera [C4357](#) Pin 4, circuit VMP19 (WH/GN), harness side and the rear video camera [C4357](#) Pin 6, circuit DMP19 (BK), harness side.



N0120904

- Measure the resistance between the rear video camera [C4357](#) Pin 3, circuit RMP19 (BN/VT), harness side and the rear video camera [C4357](#) Pin 6, circuit DMP19 (BK), harness side.



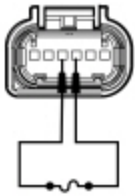
• N0120905

Are the resistances greater than 10,000 ohms?

Yes	GO to F6 .
No	REPAIR the circuit in question. TEST the system for normal operation.

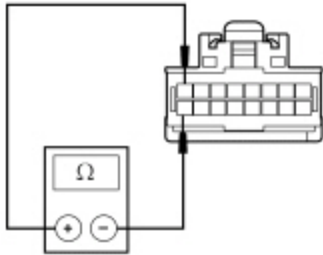
F6 CHECK REAR VIDEO SIGNAL CIRCUITS FOR AN OPEN

- Connect a fused jumper wire between the rear video camera [C4357](#) Pin 4, circuit VMP19 (WH/GN), harness side and the rear video camera [C4357](#) Pin 3, circuit RMP19 (BN/VT), harness side.



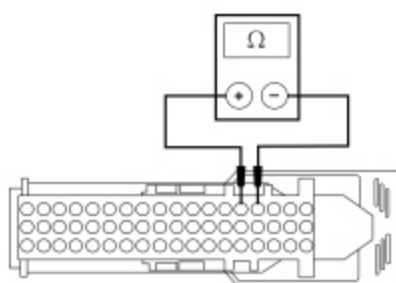
• N0120906

- For vehicles without a 203.2 mm (8 in) touchscreen, measure the resistance between the auto-dimming interior mirror [C9039](#) Pin 8, circuit VMP19 (WH/GN), harness side and the auto-dimming interior mirror [C9039](#) Pin 16, circuit RMP19 (BN/VT), harness side.



• N0069002

- For vehicles with a 203.2 mm (8 in) touchscreen, measure the resistance between the APIM [C2383](#) Pin 14, circuit VMP19 (WH/GN), harness side and the APIM [C2383](#) Pin 15, circuit RMP19 (BN/VT), harness side.



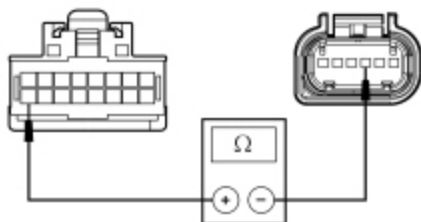
N0149434

Is the resistance less than 5 ohms?

Yes	REMOVE the jumper wire. GO to F8 .
No	REMOVE the jumper wire. GO to F7 .

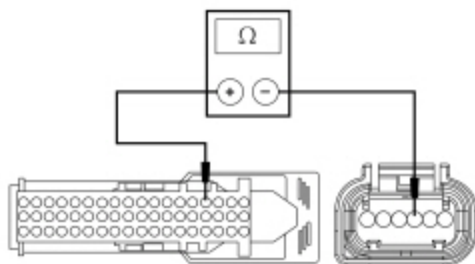
F7 CHECK REAR VIDEO (-) SIGNAL CIRCUIT FOR AN OPEN

- For vehicles without a 203.2 mm (8 in) touchscreen, measure the resistance between the auto-dimming interior mirror [C9039](#) Pin 16, circuit RMP19 (BN/VT), harness side and the rear video camera [C4357](#) Pin 3, circuit RMP19 (BN/VT), harness side.



N0120914

- For vehicles with a 203.2 mm (8 in) touchscreen, measure the resistance between the APIM [C2383](#) Pin 15, circuit RMP19 (BN/VT), harness side and the video camera [C4357](#) Pin 3, circuit RMP19 (BN/VT), harness side.



N0149436

Is the resistance less than 5 ohms?

Yes	REPAIR circuit VMP19 (WH/GN). TEST the system for normal operation.
No	REPAIR circuit RMP19 (BN/VT). TEST the system for normal operation.

F8 CHECK FOR CORRECT REAR VIDEO CAMERA OPERATION

- Connect all disconnected connectors.
- Disconnect the rear video camera connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the rear video camera connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new rear video camera. REFER to Video Camera — Rear in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test G: The Rear Video Mirror Display Is Always On

Normal Operation

With the transmission selector in REVERSE (R), a reverse signal message from the Body Control Module (BCM) is sent over a dedicated Local Interconnect Network (LIN) to the auto-dimming interior mirror, enabling the system. The video signal from the rear video camera displays an image on the left portion of the auto-dimming interior mirror display.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Auto-dimming interior mirror

PINPOINT TEST G : THE REAR VIDEO MIRROR DISPLAY IS ALWAYS ON

G1 CHECK THE REVERSING LAMPS OPERATION	
• Check the operation of the reversing lamps.	
Do the reversing lamps operate correctly?	
Yes	DISCONNECT any aftermarket trailer lighting equipment. GO to G2 .
No	REFER to Section 417-01.

G2 CHECK THE COMPASS FOR CORRECT OPERATION	
• NOTE: <i>The auto-dimming interior mirror uses the LIN for compass messaging along with messaging for the rear video camera operation.</i>	
Verify the compass operates correctly.	

Does the compass operate correctly?

Yes	GO to G3 .
No	REFER to Section 413-01 to diagnose the compass.

G3 CHECK FOR CORRECT AUTO-DIMMING INTERIOR MIRROR DISPLAY OPERATION

- Disconnect the auto-dimming interior mirror display connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the auto-dimming interior mirror display connector and make sure it seats correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	INSTALL a new auto-dimming interior mirror. REFER to Section 501-09. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test H: Visual Park Aid Is Inoperative

Refer to Wiring Diagrams Cell [131](#) , Parking Aid for schematic and connector information.

Normal Operation

When the transmission selector is in REVERSE (R) and an object is detected in the path of the parking aid sensor, a network message of the distance from the parking aid sensor to the object is sent from the Parking Aid Module (PAM) to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) . The BCM networks the parking aid sensor message to the rear video camera over the Local Interconnect Network (LIN) . The visual alerts are red, yellow or green and appear as highlights on top of the displayed image to indicate in which zone the object is detected. The visual park aid feature can be enabled and disabled within the Front Display Interface Module (FDIM) .

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Rear video camera
- PAM
- BCM

PINPOINT TEST H : VISUAL PARK AID IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

H1 VERIFY THE VISUAL PARK AID IS ENABLED

- For vehicles with a 203.2 mm (8 in) touchscreen, verify that the visual park aid is enabled in the FDIM .

- For vehicles without a 203.2 mm (8 in) touchscreen, verify that the visual park aid is enabled in the message display.

Is the visual park aid enabled in the ~~EDIM~~ or message center display?

Yes	GO to H2 .
No	ENABLE the visual park aid. TEST the system for normal operation.

H2 CHECK THE DTCS FROM THE PAM SELF-TEST

- Check the ~~PAM~~ DTCs from the self-test.

Are any ~~PAM~~ DTCs recorded?

Yes	REPAIR all PAM DTCs first. REFER to Parking Aid in this section. TEST the system for normal operation.
No	GO to H3 .

H3 CHECK THE DTCS FROM THE BCM SELF-TEST

- Check the ~~BCM~~ DTCs from the self-test.

Are any ~~BCM~~ DTCs recorded?

Yes	REFER to DTC Charts in this section.
No	GO to H4 .

H4 CHECK FOR CORRECT REAR VIDEO CAMERA OPERATION

- Connect: All Disconnected Connectors .
- Disconnect the rear video camera connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the rear video camera connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new rear video camera. REFER to Video Camera — Rear in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test I: The Front and/or Rear Video Does Not Operate Correctly — Raptor

Normal Operation — Front Video Camera

The front video camera system is active with the ignition on, the transmission selector in the drive position, the vehicle is in off road mode and the Electronic Locking Differential (ELD) or 4L activated. The front video camera receives commands from the Body Control Module (BCM) over a dedicated Local Interconnect Network (LIN) . The front video camera sends a video signal to the camera switching module on a pair of shielded video circuits. The Front Display Interface Module (FDIM) receives the signal from the camera switching module, the **FDIM** displays the video image of the area in front of the vehicle.

Normal Operation — Rear Video Camera

The rear video camera system is active with the ignition on and the transmission selector in the reverse position. The rear video camera receives commands from the **BCM** over a dedicated **LIN** . The rear video camera sends a video signal to the camera switching module on a pair of shielded video circuits. The **FDIM** receives the signal from the camera switching module, the **FDIM** displays the video image of the area behind the vehicle.

- DTC B115E:08 (Camera Module: Bus Signal/Message Failure) — a continuous and a on-demand DTC that sets in the **BCM** when there is a **LIN** communication fault from the rear video camera.
- DTC C1001:01 (Vision System Camera: General Electrical Failure) — a continuous DTC that sets in the **ACM** when no video signal is detected from the rear video camera when the vehicle is in reverse.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Front video camera
- Rear video camera
- Camera switching module
- **FDIM**
- **BCM**

PINPOINT TEST I : THE FRONT AND/OR REAR VIDEO DOES NOT OPERATE CORRECTLY — RAPTOR

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.

I1 CHECK THE OPERATION OF THE FRONT AND REAR VIDEO CAMERAS

- Check the operation of the front video camera system and verify the front video camera image.
- Check the operation of the rear video camera system and verify the rear video camera image.

Are both cameras inoperative?

Yes	GO to I2 .
No	For the front video camera, GO to I17 . For the rear video camera, GO to I3 .

I2 CHECK THE DTCS FROM THE BCM SELF-TEST

- Check the **BCM** DTCs from the self-test.

Are any **BCM** DTCs retrieved?

Yes	For BCM DTC B115E:08, GO to I37 . For all other DTCs, REFER to DTC Charts in this section.
No	GO to I31 .

I3 CHECK AND CLEAN THE REAR VIDEO CAMERA LENS

- Clean and make sure the rear video camera lens is clear of any debris.
- Test the rear video camera system and verify the rear video camera image.

Is the rear video camera concern still present?

Yes	GO to I4 .
No	The system is operating correctly at this time. The concern was caused by a dirty rear video camera lens.

I4 CHECK THE REVERSING LAMPS OPERATION

- Apply the parking brake.
- Ignition ON.
- Select REVERSE.
- Check the operation of the reversing lamps.

Do the reversing lamps operate correctly?

Yes	GO to I5 .
No	REFER to Section 417-01.

I5 VERIFY AND ENABLE THE INOPERATIVE REAR VIDEO CAMERA FEATURE

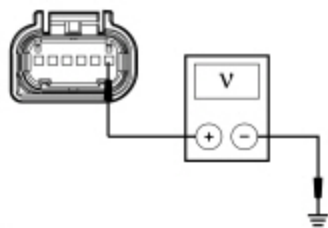
- Verify and enable the inoperative rear video camera feature within the **EDIM** (with a 203.2 mm (8 in) touchscreen) or message center (without a 203.2 mm (8 in) touchscreen).

Does the rear video camera feature operate correctly?

Yes	The rear video camera system is operating correctly at this time. The rear video camera feature concern was caused by the feature being disabled. INSTRUCT the customer on how to enable the rear video camera features. REFER to the Owner's Literature.
No	GO to I6 .

I6 CHECK THE REAR VIDEO CAMERA VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Rear Video Camera [C4357](#).
- Ignition ON.
- Measure the voltage between the rear video camera [C4357](#) Pin 1, circuit CBP34 (VT/BN), harness side and ground.



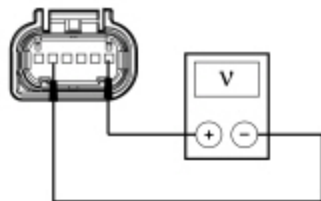
N0120901

Is the voltage greater than 10 volts?

Yes	GO to 17 .
No	REPAIR the circuit.

I7 CHECK THE REAR VIDEO CAMERA GROUND CIRCUIT FOR AN OPEN

- Measure the voltage between the rear video camera [C4357](#) Pin 1, circuit CBP34 (VT/BN), harness side and the rear video camera [C4357](#) Pin 5, circuit GD120 (BK/GN), harness side and ground.



N0120902

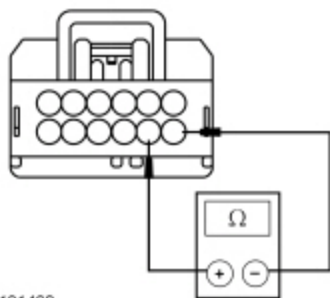
Is the voltage greater than 10 volts?

Yes	GO to 18 .
No	REPAIR circuit GD120 (BK/GN) for an open. TEST the system for normal operation.

I8 CHECK THE REAR VIDEO SIGNAL CIRCUITS INTEGRITY

- Ignition OFF.
- Disconnect: Camera Switching Module [C1664](#).

- Connect: Rear Video Camera [C4357](#).
- Measure the resistance between the camera switching module [C1664](#) Pin 7, circuit VMP19 (WH), harness side and the camera switching module [C1664](#) Pin 8, circuit RMP19 (BN), harness side.



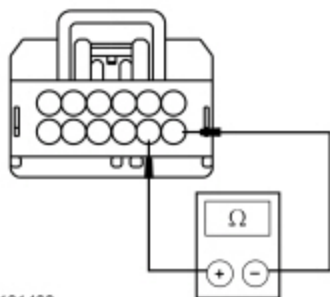
• N0131468

Is the resistance between 74 and 106 ohms?

Yes	GO to I10 .
No	If the resistance is less than 74 ohms, GO to I9 . If the resistance is greater than 106 ohms, GO to I13 .

I9 CHECK THE REAR VIDEO SIGNAL CIRCUITS FOR A SHORT TOGETHER

- Disconnect: Rear Video Camera [C4357](#).
- Measure the resistance between the camera switching module [C1664](#) Pin 7, circuit VMP19 (WH), harness side and the camera switching module [C1664](#) Pin 8, circuit RMP19 (BN), harness side.



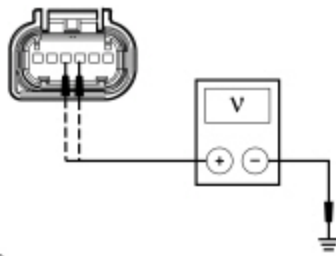
• N0131468

Is the resistance less than 10,000 ohms?

Yes	REPAIR the circuits.
No	GO to I16 .

I10 CHECK THE REAR VIDEO CAMERA VIDEO SIGNAL CIRCUITS FOR A SHORT TO VOLTAGE

- Disconnect: Rear Video Camera [C4357](#).
- Ignition ON.
- Measure the voltage between the rear video camera [C4357](#) Pin 4, circuit VMP19 (WH/GN), harness side and ground; and between the rear video camera [C4357](#) Pin 3, circuit RMP19 (BN/VT), harness side and ground.



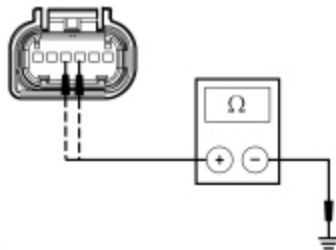
N0120909

Is any voltage present?

Yes	REPAIR the circuits. TEST the system for normal operation.
No	GO to I11 .

I11 CHECK THE REAR VIDEO CAMERA SIGNAL CIRCUITS FOR A SHORT TO GROUND

- Measure the resistance between the rear video camera [C4357](#) Pin 4, circuit VMP19 (WH/GN), harness side and ground; and between the rear video camera [C4357](#) Pin 3, circuit RMP19 (BN/VT), harness side and ground.



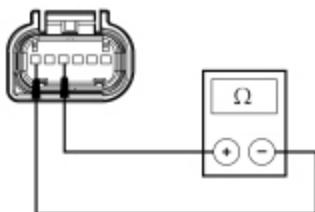
N0121053

Are the resistances greater than 10,000 ohms?

Yes	GO to I12 .
No	REPAIR the circuit in question. TEST the system for normal operation.

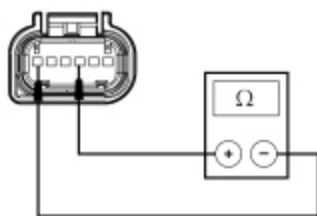
I12 CHECK THE REAR VIDEO CAMERA VIDEO SIGNAL CIRCUITS FOR A SHORT TO VIDEO SHIELD

- Measure the resistance between the rear video camera [C4357](#) Pin 4, circuit VMP19 (WH/GN), harness side and the rear video camera [C4357](#) Pin 6, circuit DMP19 (BK), harness side.



N0120904

- Measure the resistance between the rear video camera [C4357](#) Pin 3, circuit RMP19 (BN/VT), harness side and the rear video camera [C4357](#) Pin 6, circuit DMP19 (BK), harness side.



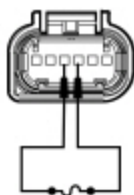
• N0120905

Are the resistances greater than 10,000 ohms?

Yes	GO to I15 .
No	REPAIR the circuit in question. TEST the system for normal operation.

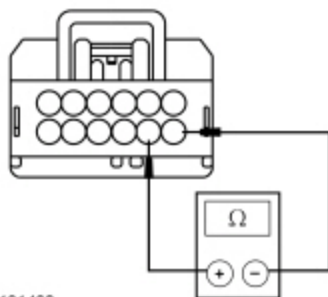
I13 CHECK REAR VIDEO SIGNAL CIRCUITS FOR AN OPEN

- Ignition OFF.
- Connect a fused jumper wire between the rear video camera [C4357](#) Pin 4, circuit VMP19 (WH/GN), harness side and the rear video camera [C4357](#) Pin 3, circuit RMP19 (BN/VT), harness side.



• N0120906

- Measure the resistance between the camera switching module [C1664](#) Pin 7, circuit VMP19 (WH), harness side and the camera switching module [C1664](#) Pin 8, circuit RMP19 (BN), harness side.



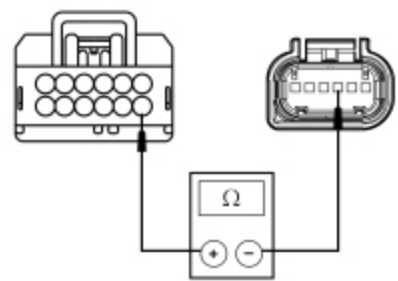
• N0131468

Is the resistance less than 5 ohms?

Yes	REMOVE the jumper wire. GO to I16 .
No	REMOVE the jumper wire. GO to I14 .

I14 CHECK REAR VIDEO (+) SIGNAL CIRCUIT FOR AN OPEN

- Measure the resistance between the camera switching module [C1664](#) Pin 7, circuit VMP19 (WH), harness side and the rear video camera [C4357](#) Pin 4, circuit VMP19 (WH/GN), harness side.



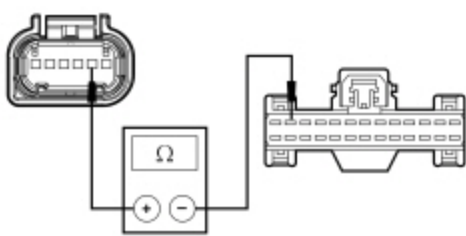
• N0131469

Is the resistance less than 5 ohms?

Yes	REPAIR circuit RMP19. TEST the system for normal operation.
No	REPAIR circuit VMP19. TEST the system for normal operation.

I15 CHECK THE LIN CIRCUIT FOR AN OPEN

- Disconnect: Front Video Camera [C1663](#).
- Disconnect: BCM [C2280C](#) .
- Measure the resistance between the rear video camera [C4357](#) Pin 2, circuit VDN03 (GY/VT), harness side and the BCM [C2280C](#) Pin 12, circuit VDN03 (GY/VT), harness side.



• N0177798

Is resistance less than 5 ohms between the rear video camera and BCM ?

Yes	GO to I16 .
No	REPAIR the circuit. TEST the system for normal operation.

I16 CHECK FOR CORRECT REAR VIDEO CAMERA OPERATION

- Connect: All Disconnected Connectors .
- Disconnect the rear video camera connector.
- Check for:

- corrosion
- damaged pins
- pushed-out pins

- Connect the rear video camera connector and make sure it seats correctly.
- Verify the rear video camera is configured correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new rear video camera. REFER to Video Camera — Rear in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

I17 CHECK AND CLEAN THE FRONT VIDEO CAMERA LENS

- Clean and make sure the front video camera lens is clear of any debris.
- Test the front video camera system and verify the front video camera image.

Is the front video camera concern still present?

Yes	GO to I18 .
No	The system is operating correctly at this time. The concern was caused by a dirty front video camera lens.

I18 CHECK THE 4X4 LOW OPERATION

- Ignition ON.
- Shift the transfer case to 4L (4X4 LOW).
- Select DRIVE.

Does the 4L engage correctly?

Yes	GO to I19 .
No	REFER to Section 308-07A.

I19 VERIFY AND ENABLE THE INOPERATIVE FRONT VIDEO CAMERA FEATURE

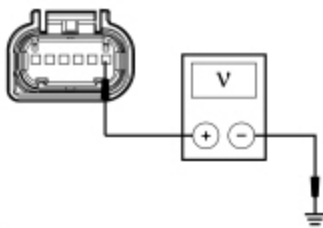
- Verify and enable the inoperative front video camera feature within the **EDIM** .

Does the front video camera feature operate correctly?

Yes	The front video camera system is operating correctly at this time. The front video camera feature concern was caused by the feature being disabled. INSTRUCT the customer on how to enable the front video camera features, REFER to the Owner's Literature.
No	GO to I20 .

I20 CHECK THE FRONT VIDEO CAMERA VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Front Video Camera [C1663](#).
- Ignition ON.
- Measure the voltage between the front video camera [C1663](#) Pin 1, circuit CBP34 (VT/BN), harness side and ground.



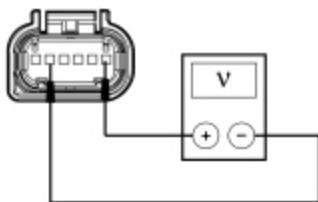
N0120901

Is the voltage greater than 10 volts?

Yes	GO to I21 .
No	REPAIR the circuit.

I21 CHECK THE FRONT VIDEO CAMERA GROUND CIRCUIT FOR AN OPEN

- Measure the voltage between the front video camera [C1663](#) Pin 1, circuit CBP34 (VT/BN), harness side and the front video camera [C1663](#) Pin 5, circuit GD120 (BK/GN), harness side.



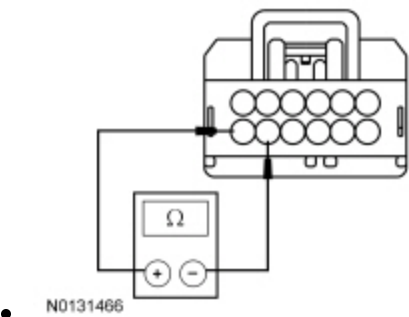
N0120902

Is the voltage greater than 10 volts?

Yes	GO to I22 .
No	REPAIR circuit GD120 (BK/GN) for an open. TEST the system for normal operation.

I22 CHECK THE FRONT VIDEO SIGNAL CIRCUITS INTEGRITY

- Ignition OFF.
- Connect: Front Video Camera [C1663](#).
- Disconnect: Camera Switching Module [C1664](#).
- Measure the resistance between the camera switching module [C1664](#) Pin 12, circuit VMP19 (WH/GN), harness side and the camera switching module [C1664](#) Pin 11, circuit RMP19 (BN/VT), harness side.

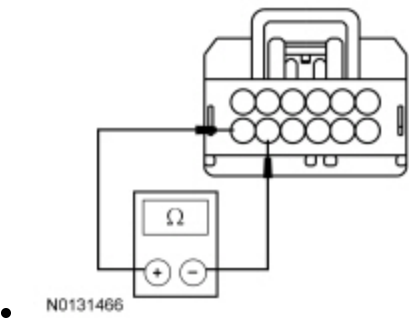


Is the resistance between 74 and 106 ohms?

Yes	GO to I24 .
No	If less than 74 ohms, GO to I23 . If greater than 106 ohms, GO to I27 .

I23 CHECK THE FRONT VIDEO SIGNAL CIRCUITS FOR A SHORT TOGETHER

- Disconnect: Front Video Camera [C1663](#).
- Measure the resistance between the camera switching module [C1664](#) Pin 12, circuit VMP19 (WH/GN), harness side and the camera switching module [C1664](#) Pin 11, circuit RMP19 (BN/VT), harness side.



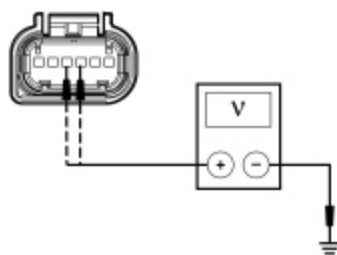
Is the resistance the greater than 10,000 ohms?

Yes	GO to I30 .
No	REPAIR the circuits.

I24 CHECK THE FRONT VIDEO CAMERA VIDEO SIGNAL CIRCUITS FOR A SHORT TO VOLTAGE

- Disconnect: Front Video Camera [C1663](#).

- Measure the voltage between the front video camera [C1663](#) Pin 4, circuit VMP19 (WH/GN), harness side and ground; and between the front video camera [C1663](#) Pin 3, circuit RMP19 (BN/VT), harness side and ground.



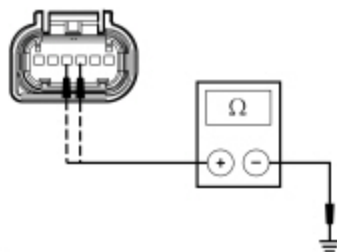
N0120909

Is any voltage present?

Yes	REPAIR the circuit in question. TEST the system for normal operation.
No	GO to I25 .

I25 CHECK THE FRONT VIDEO CAMERA SIGNAL CIRCUITS FOR A SHORT TO GROUND

- Disconnect: Front Video Camera [C1663](#).
- Measure the resistance between the front video camera [C1663](#) Pin 4, circuit VMP19 (WH/GN), harness side and ground; and between the front video camera [C1663](#) Pin 3, circuit RMP19 (BN/VT), harness side and ground.



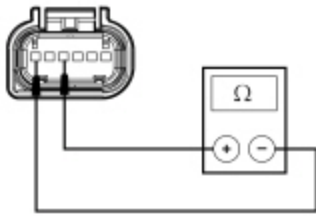
N0121053

Are the resistances greater than 10,000 ohms?

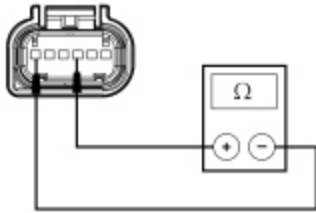
Yes	GO to I26 .
No	REPAIR the circuit in question. TEST the system for normal operation.

I26 CHECK THE FRONT VIDEO CAMERA VIDEO SIGNAL CIRCUITS FOR A SHORT TO VIDEO SHIELD

- Measure the resistance between the front video camera [C1663](#) Pin 4, circuit VMP19 (WH/GN), harness side and the front video camera [C1663](#) Pin 6, circuit DMP19 (BK), harness side.



- N0120904
- Measure the resistance between the front video camera [C1663](#) Pin 3, circuit RMP19 (BN/VT), harness side and the front video camera [C1663](#) Pin 6, circuit DMP19 (BK), harness side.



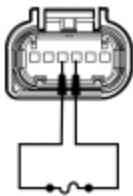
- N0120905

Are the resistances greater than 10,000 ohms?

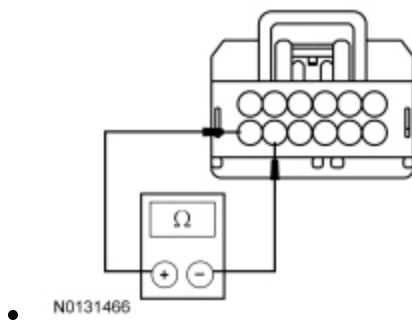
Yes	GO to I29 .
No	REPAIR the circuit in question. TEST the system for normal operation.

I27 CHECK FRONT VIDEO SIGNAL CIRCUITS FOR AN OPEN

- Ignition OFF.
- Connect a fused jumper wire between the front video camera [C1663](#) Pin 4, circuit VMP19 (WH/GN), harness side and the front video camera [C1663](#) Pin 3, circuit RMP19 (BN/VT), harness side.



- N0120906
- Measure the resistance between the camera switching module [C1664](#) Pin 12, circuit VMP19 (WH/GN), harness side and the camera switching module [C1664](#) Pin 11, circuit RMP19 (BN/VT), harness side.

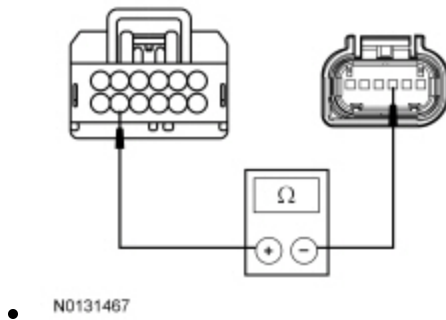


Is the resistance less than 5 ohms?

Yes	REMOVE the jumper wire. GO to I30 .
No	REMOVE the jumper wire. GO to I28 .

I28 CHECK FRONT VIDEO (+) SIGNAL CIRCUIT FOR AN OPEN

- Measure the resistance between the camera switching module [C1664](#) Pin 11, circuit RMP19 (BN/VT), harness side and the front video camera [C1663](#) Pin 3, circuit RMP19 (BN/VT), harness side.

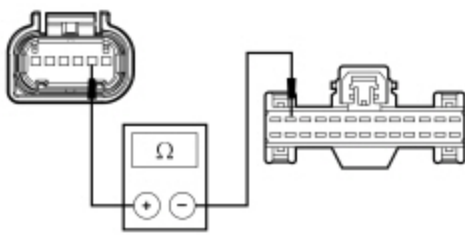


Is the resistance less than 5 ohms?

Yes	REPAIR circuit VMP19 (WH/GN). TEST the system for normal operation.
No	REPAIR circuit RMP19 (BN/VT). TEST the system for normal operation.

I29 CHECK THE LIN CIRCUIT FOR AN OPEN

- Disconnect: Rear Video Camera [C4357](#).
- Disconnect: BCM [C2280C](#) .
- Measure the resistance between the front video camera [C1663](#) Pin 2, circuit VDN03 (GY/VT), harness side and the BCM [C2280C](#) Pin 12, circuit VDN03 (GY/VT), harness side; and between the front video camera [C1663](#) Pin 2, circuit VDN03 (GY/VT), harness side and ground.



N0177798

Is resistance less than 5 ohms between the front video camera and BCM ; and greater than 10,000 ohms between the front video camera and ground?

Yes	GO to I30 .
No	REPAIR the circuit. TEST the system for normal operation.

I30 CHECK FOR CORRECT FRONT VIDEO CAMERA OPERATION

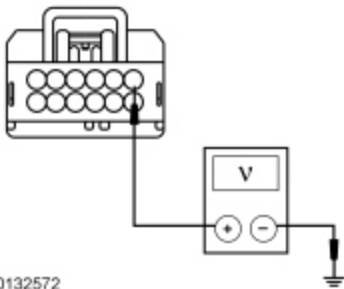
- Connect: All Disconnected Connectors .
- Disconnect the front video camera connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the front video camera connector and make sure it seats correctly.
- Verify the front video camera is configured correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new front video camera. REFER to Video Camera — Front in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

I31 CHECK THE CAMERA SWITCHING MODULE POWER CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Camera Switching Module [C1664](#).
- Ignition ON.
- Measure the voltage between the camera switching module [C1664](#) Pin 1, circuit CBP34 (VT/BN), harness side and ground.



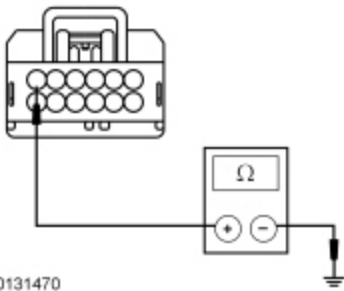
• N0132572

Is the voltage greater than 10 volts?

Yes	GO to I32 .
No	REPAIR the circuit. TEST the system for normal operation.

I32 CHECK THE CAMERA SWITCHING MODULE GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the camera switching module [C1664](#) Pin 6, circuit GD120 (BK/GN), harness side and ground.



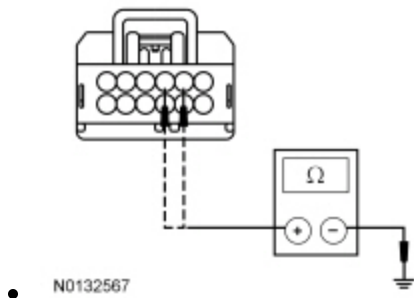
• N0131470

Is the resistance less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to I33 .
No	REPAIR the circuit. CONNECT the negative battery cable. TEST the system for normal operation.

I33 CHECK THE VIDEO SIGNAL CIRCUITS FOR A SHORT TO GROUND

- Disconnect: APIM [C2383](#) .
- Measure the resistance between the camera switching module [C1664](#) Pin 2, circuit VMP19 (WH/GN), harness side and ground; and between the camera switching module [C1664](#) Pin 3, circuit RMP19 (BN/VT), harness side and ground.

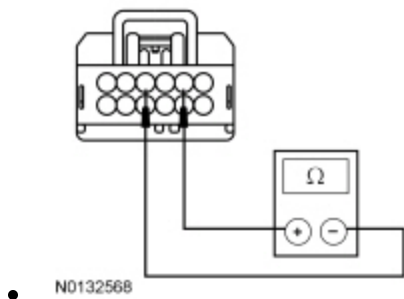


Are the resistances greater than 10,000 ohms?

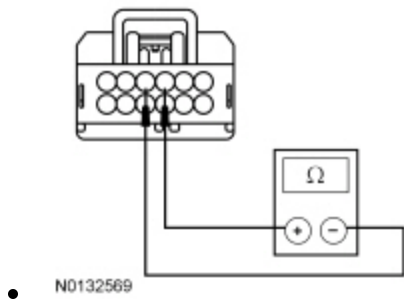
Yes	GO to I34 .
No	REPAIR the circuit in question. TEST the system for normal operation.

I34 CHECK THE VIDEO SIGNAL CIRCUITS FOR A SHORT TO VIDEO SHIELD

- Measure the resistance between the camera switching module [C1664](#) Pin 2, circuit VMP19 (WH/GN), harness side and the camera switching module [C1664](#) Pin 4, circuit DMP19 (BK), harness side.



- Measure the resistance between the camera switching module [C1664](#) Pin 3, circuit RMP19 (BN/VT), harness side and the camera switching module [C1664](#) Pin 4, circuit DMP19 (BK), harness side.

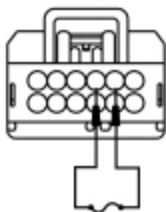


Are the resistances greater than 10,000 ohms?

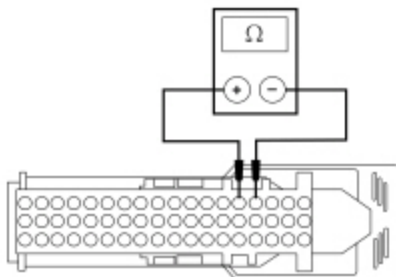
Yes	GO to I35 .
No	REPAIR the circuit in question. TEST the system for normal operation.

I35 CHECK VIDEO SIGNAL CIRCUITS FOR AN OPEN BETWEEN THE CAMERA SWITCHING MODULE AND THE APIM

- Connect a fused jumper wire between the camera switching module [C1664](#) Pin 2, circuit VMP19 (WH/GN), harness side and the camera switching module [C1664](#) Pin 3, circuit RMP19 (BN/VT), harness side.



- N0132571
- Measure the resistance between the APIM [C2383](#) Pin 14, circuit VMP19 (WH/GN), harness side and the APIM [C2383](#) Pin 15, circuit RMP19 (BN/VT), harness side.



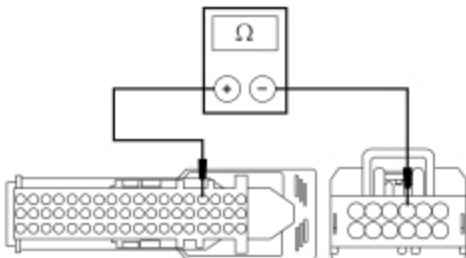
• N0149434

Is the resistance less than 5 ohms?

Yes	REMOVE the jumper wire. GO to I37 .
No	REMOVE the jumper wire. GO to I36 .

I36 CHECK VIDEO (-) SIGNAL CIRCUIT FOR AN OPEN BETWEEN THE CAMERA SWITCHING MODULE AND THE APIM

- Measure the resistance between the camera switching module [C1664](#) Pin 3, circuit RMP19 (BN/VT), harness side and the APIM [C2383](#) Pin 15, circuit RMP19 (BN/VT), harness side.



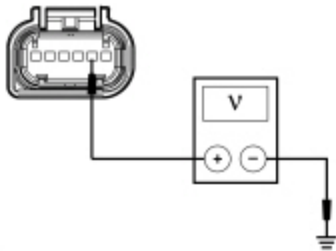
• N0149437

Is the resistance less than 5 ohms?

Yes	REPAIR circuit VMP19 (WH/GN). TEST the system for normal operation.
No	REPAIR circuit RMP19 (BN/VT). TEST the system for normal operation.

I37 CHECK THE LIN CIRCUIT FOR A SHORT TO VOLTAGE

- Disconnect: Front Video Camera [C1663](#).
- Disconnect: Rear Video Camera [C4357](#).
- Disconnect: BCM [C2280C](#) .
- Ignition ON.
- Measure the voltage between the rear video camera [C4357](#) Pin 2, circuit VDN03 (GY/VT), harness side and ground.



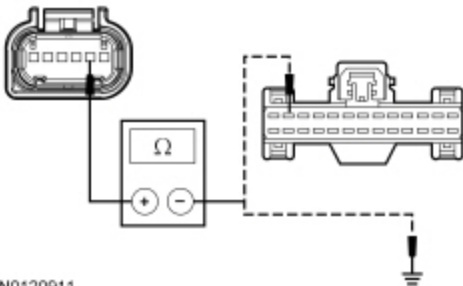
N0120910

Is any voltage present?

Yes	REPAIR the circuit. TEST the system for normal operation.
No	GO to I38 .

I38 CHECK THE LIN CIRCUIT FOR AN OPEN OR SHORT TO GROUND

- Ignition OFF.
- Measure the resistance between the rear video camera [C4357](#) Pin 2, circuit VDN03 (GY/VT), harness side and the BCM [C2280C](#) Pin 12, circuit VDN03 (GY/VT), harness side; and between the rear video camera [C4357](#) Pin 2, circuit VDN03 (GY/VT), harness side and ground.



N0120911

Is the resistance less than 5 ohms between the rear video camera and the BCM ; and greater than 10,000 ohms between the rear video camera and ground?

Yes	GO to I39 .
No	REPAIR the circuit. TEST the system for normal operation.

I39 CHECK FOR CORRECT REAR VIDEO CAMERA OPERATION

- Connect all disconnected connectors.
- Disconnect the rear video camera connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the rear video camera connector and make sure it seats correctly.
- Verify the rear video camera is configured correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	REPLACE the rear video camera. Video Camera — Rear If the concern is still present after the rear video camera replacement, GO to I40 .
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

I40 CHECK FOR CORRECT FRONT VIDEO CAMERA OPERATION

- Connect all disconnected connectors.
- Disconnect the camera switching module connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the camera switching module connector and make sure it seats correctly.
- Verify the front video camera is configured correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	REPLACE the front video camera. Video Camera — Front If the concern is still present after the front video camera replacement, GO to I41 .
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

I41 CHECK FOR CORRECT CAMERA SWITCHING MODULE OPERATION

- Connect all disconnected connectors.
- Disconnect the camera switching module connector.
- Check for:
 - corrosion

- damaged pins
- pushed-out pins

- Connect the camera switching module connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new camera switching module. REFER to Camera Switching Module in this section. TEST the system for normal operation. If the concern is still present after the camera switching module replacement, GO to I42 .
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

I42 CHECK FOR CORRECT BCM OPERATION

- Ignition OFF.
- Disconnect and inspect all BCM connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed out pins (Install new pins as necessary)
- Reconnect the BCM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test J: Poor Image Quality — Raptor

Normal Operation — Front Video Camera

The front video camera system is active with the ignition on, the transmission selector in the drive position, the vehicle is in off road mode and the Electronic Locking Differential (ELD) or 4L activated. The front video camera receives commands from the Body Control Module (BCM) over a dedicated Local Interconnect Network (LIN) . The front video camera sends a video signal to the camera switching module on a pair of shielded video circuits. The Front Display Interface Module (FDIM) receives the signal from the camera switching module and displays the video image of the area in front of the vehicle.

Normal Operation — Rear Video Camera

The rear video camera system is active with the ignition on and the transmission selector in the reverse position. The rear video camera receives commands from the BCM over a dedicated LIN . The rear video camera sends a video signal to the camera switching module on a pair of shielded video circuits. The FDM receives the signal from the camera switching module and displays the video image of the area behind the vehicle.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Front video camera
- Rear video camera
- Camera switching module

PINPOINT TEST J : POOR IMAGE QUALITY – RAPTOR

NOTE: Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.

J1 CHECK THE OPERATION OF THE FRONT AND REAR VIDEO CAMERAS

- Check the operation of the front video camera system and verify the front video camera image.
- Check the operation of the rear video camera system and verify the rear video camera image.

Do both cameras have poor image quality?

Yes	GO to J18.
No	For the front video camera, GO to J2. For the rear video camera, GO to J10.

J2 CHECK AND CLEAN THE FRONT VIDEO CAMERA LENS

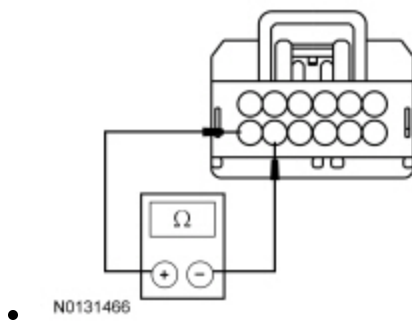
- Clean and make sure the front video camera lens is clear of any debris.
- Test the front video camera system and verify the front video camera image.

Is the front video camera concern still present?

Yes	GO to J3.
No	The system is operating correctly at this time. The concern was caused by a dirty front video camera lens.

J3 CHECK THE FRONT VIDEO SIGNAL CIRCUITS INTEGRITY

- Ignition OFF.
- Disconnect: Camera Switching Module C1664.
- Measure the resistance between the camera switching module C1664 Pin 12, circuit VMP19 (WH/GN), harness side and the camera switching module C1664 Pin 11, circuit RMP19 (BN/VT), harness side.

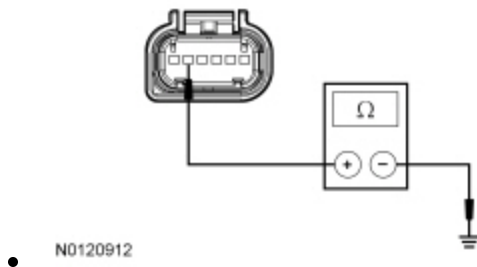


Is the resistance between 74 and 106 ohms?

Yes	GO to J4 .
No	GO to J5 .

J4 CHECK THE FRONT VIDEO CAMERA GROUND CIRCUIT FOR AN OPEN

- Disconnect: Negative Battery Cable .
- Disconnect: Front Video Camera [C1663](#).
- Measure the resistance between the front video camera [C1663](#) Pin 5, circuit GD120 (BK/GN), harness side and ground.

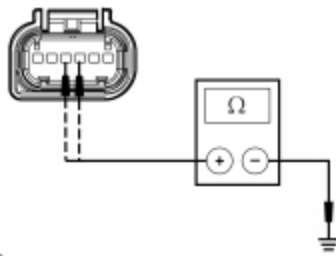


Is the resistance less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to J5 .
No	REPAIR the circuit. CONNECT the negative battery cable. TEST the system for normal operation.

J5 CHECK THE FRONT VIDEO CAMERA SIGNAL CIRCUITS FOR A SHORT TO GROUND

- Measure the resistance between the front video camera [C1663](#) Pin 4, circuit VMP19 (WH/GN), harness side and ground; and between the front video camera [C1663](#) Pin 3, circuit RMP19 (BN/VT), harness side and ground.



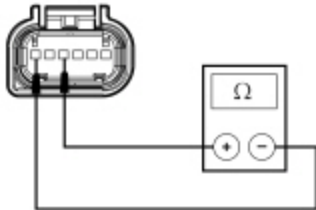
N0121053

Are the resistances greater than 10,000 ohms?

Yes	GO to J6 .
No	REPAIR the circuit in question. TEST the system for normal operation.

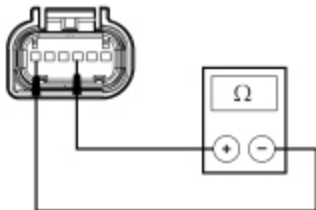
J6 CHECK THE FRONT VIDEO CAMERA VIDEO SIGNAL CIRCUITS FOR A SHORT TO VIDEO SHIELD

- Measure the resistance between the front video camera [C1663](#) Pin 4, circuit VMP19 (WH/GN), harness side and the front video camera [C1663](#) Pin 6, circuit DMP19 (BK), harness side.



N0120904

- Measure the resistance between the front video camera [C1663](#) Pin 3, circuit RMP19 (BN/VT), harness side and the front video camera [C1663](#) Pin 6, circuit DMP19 (BK), harness side.



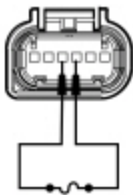
N0120905

Are the resistances greater than 10,000 ohms?

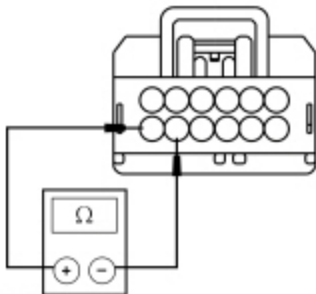
Yes	GO to J7 .
No	REPAIR the circuit in question. TEST the system for normal operation.

J7 CHECK FRONT VIDEO SIGNAL CIRCUITS FOR AN OPEN

- Connect a fused jumper wire between the front video camera [C1663](#) Pin 4, circuit VMP19 (WH/GN), harness side and the front video camera [C1663](#) Pin 3, circuit RMP19 (BN/VT), harness side.



- N0120906
- Measure the resistance between the camera switching module [C1664](#) Pin 12, circuit VMP19 (WH/GN), harness side and the camera switching module [C1664](#) Pin 11, circuit RMP19 (BN/VT), harness side.



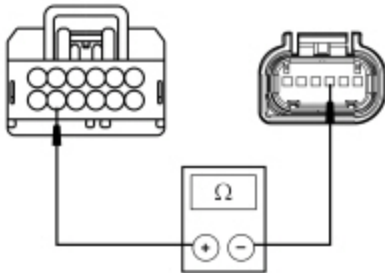
• N0131466

Is the resistance less than 5 ohms?

Yes	REMOVE the jumper wire. GO to J9 .
No	REMOVE the jumper wire. GO to J8 .

J8 CHECK FRONT VIDEO (-) SIGNAL CIRCUIT FOR AN OPEN

- Measure the resistance between the camera switching module [C1664](#) Pin 11, circuit RMP19 (BN/VT), harness side and the front video camera [C1663](#) Pin 3, circuit RMP19 (BN/VT), harness side.



• N0131467

Is the resistance less than 5 ohms?

Yes	REPAIR circuit VMP19 (WH/GN). TEST the system for normal operation.
No	REPAIR circuit RMP19 (BN/VT). TEST the system for normal operation.

J9 CHECK FOR CORRECT FRONT VIDEO CAMERA OPERATION

- Connect all disconnected connectors.
- Disconnect the front video camera connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the front video camera connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new front video camera. REFER to Video Camera — Front in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

J10 CHECK AND CLEAN THE REAR VIDEO CAMERA LENS

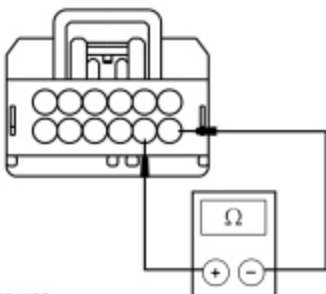
- Clean and make sure the rear video camera lens is clear of any debris.
- Test the rear video camera system and verify the rear video camera image.

Is the rear video camera concern still present?

Yes	GO to J11 .
No	The system is operating correctly at this time. The concern was caused by a dirty rear video camera lens.

J11 CHECK THE REAR VIDEO SIGNAL CIRCUITS INTEGRITY

- Ignition OFF.
- Disconnect: Camera Switching Module [C1664](#).
- Measure the resistance between the camera switching module [C1664](#) Pin 7, circuit VMP19 (WH/GN), harness side and the camera switching module [C1664](#) Pin 8, circuit RMP19 (BN/VT), harness side.



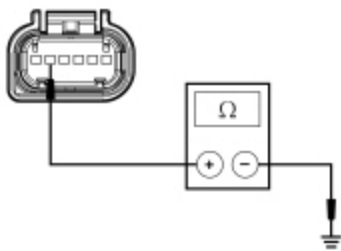
- N0131468

Is the resistance between 74 and 106 ohms?

Yes	GO to J12 .
No	GO to J13 .

J12 CHECK THE REAR VIDEO CAMERA GROUND CIRCUIT FOR AN OPEN

- Disconnect: Negative Battery Cable .
- Disconnect: Rear Video Camera [C4357](#).
- Measure the resistance between the rear video camera [C4357](#) Pin 5, circuit GD120 (BK/GN), harness side and ground.

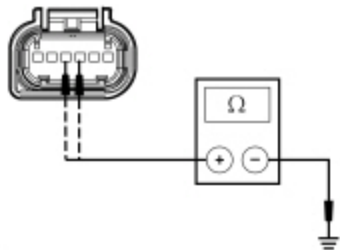


Is the resistance less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to J13 .
No	REPAIR the circuit. CONNECT the negative battery cable. TEST the system for normal operation.

J13 CHECK THE REAR VIDEO CAMERA SIGNAL CIRCUITS FOR A SHORT TO GROUND

- Measure the resistance between the rear video camera [C4357](#) Pin 4, circuit VMP19 (WH/GN), harness side and ground; and between the rear video camera [C4357](#) Pin 3, circuit RMP19 (BN/VT), harness side and ground.

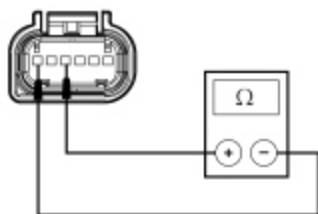


Are the resistances greater than 10,000 ohms?

Yes	GO to J14 .
No	REPAIR the circuit in question. TEST the system for normal operation.

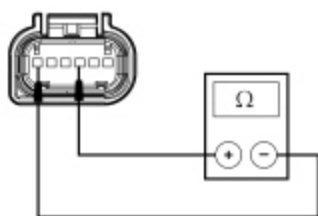
J14 CHECK THE REAR VIDEO CAMERA VIDEO SIGNAL CIRCUITS FOR A SHORT TO VIDEO SHIELD

- Measure the resistance between the rear video camera [C4357](#) Pin 4, circuit VMP19 (WH/GN), harness side and the rear video camera [C4357](#) Pin 6, circuit DMP19 (BK), harness side.



• N0120904

- Measure the resistance between the rear video camera [C4357](#) Pin 3, circuit RMP19 (BN/VT), harness side and the rear video camera [C4357](#) Pin 6, circuit DMP19 (BK), harness side.



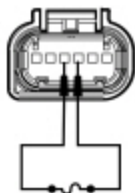
• N0120905

Are the resistances greater than 10,000 ohms?

Yes	GO to J15 .
No	REPAIR the circuit in question. TEST the system for normal operation.

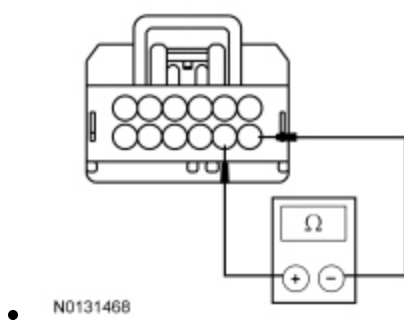
J15 CHECK REAR VIDEO SIGNAL CIRCUITS FOR AN OPEN

- Connect a fused jumper wire between the rear video camera [C4357](#) Pin 4, circuit VMP19 (WH/GN), harness side and the rear video camera [C4357](#) Pin 3, circuit RMP19 (BN/VT), harness side.



• N0120906

- Measure the resistance between the camera switching module [C1664](#) Pin 7, circuit VMP19 (WH/GN), harness side and the camera switching module [C1664](#) Pin 8, circuit RMP19 (BN/VT), harness side.

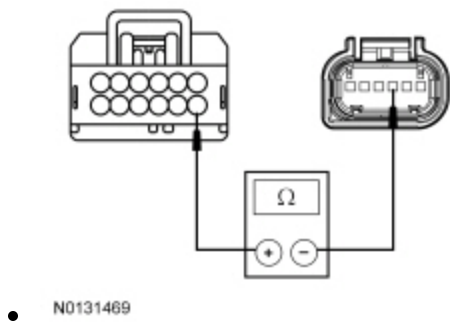


Is the resistance less than 5 ohms?

Yes	REMOVE the jumper wire. GO to J17 .
No	REMOVE the jumper wire. GO to J16 .

J16 CHECK REAR VIDEO (-) SIGNAL CIRCUIT FOR AN OPEN

- Measure the resistance between the camera switching module [C1664](#) Pin 8, circuit RMP19 (BN/VT), harness side and the rear video camera [C4357](#) Pin 3, circuit RMP19 (BN/VT), harness side.



Is the resistance less than 5 ohms?

Yes	REPAIR circuit VMP19 (WH/GN). TEST the system for normal operation.
No	REPAIR circuit RMP19 (BN/VT). TEST the system for normal operation.

J17 CHECK FOR CORRECT REAR VIDEO CAMERA OPERATION

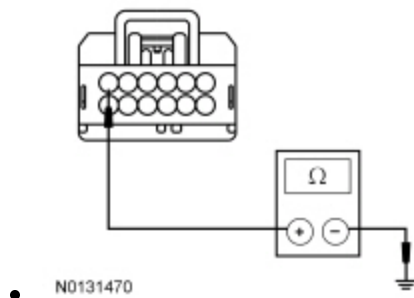
- Connect all disconnected connectors.
- Disconnect the rear video camera connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the rear video camera connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new rear video camera. REFER to Video Camera — Rear in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

J18 CHECK THE CAMERA SWITCHING MODULE GROUND CIRCUIT FOR AN OPEN

- Disconnect: Negative Battery Cable .
- Disconnect: Camera Switching Module [C1664](#).
- Measure the resistance between the camera switching module [C1664](#) Pin 6, circuit GD120 (BK/GN), harness side and ground.

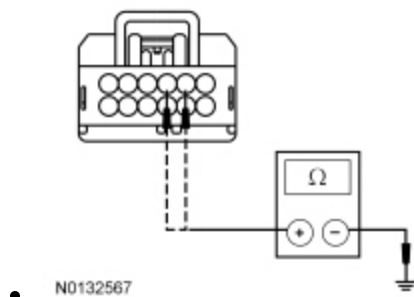


Is the resistance less than 5 ohms?

Yes	CONNECT the negative battery cable. GO to J19 .
No	REPAIR the circuit. CONNECT the negative battery cable. TEST the system for normal operation.

J19 CHECK THE VIDEO SIGNAL CIRCUITS FOR A SHORT TO GROUND

- Disconnect: APIM [C2383](#) .
- Measure the resistance between the camera switching module [C1664](#) Pin 2, circuit VMP19 (WH/GN), harness side and ground; and between the camera switching module [C1664](#) Pin 3, circuit RMP19 (BN/VT), harness side and ground.

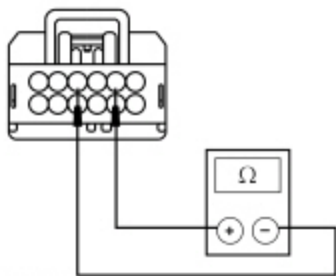


Are the resistances greater than 10,000 ohms?

Yes	GO to J20 .
No	REPAIR the circuit in question. TEST the system for normal operation.

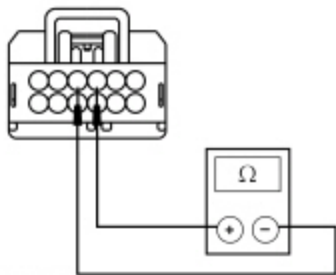
J20 CHECK THE VIDEO CAMERA VIDEO SIGNAL CIRCUITS FOR A SHORT TO VIDEO SHIELD

- Measure the resistance between the camera switching module [C1664](#) Pin 2, circuit VMP19 (WH/GN), harness side and the camera switching module [C1664](#) Pin 4, circuit DMP19 (BK), harness side.



• N0132568

- Measure the resistance between the camera switching module [C1664](#) Pin 3, circuit RMP19 (BN/VT), harness side and the camera switching module [C1664](#) Pin 4, circuit DMP19 (BK), harness side.



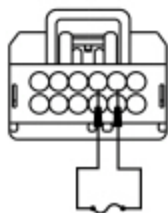
• N0132569

Are the resistances greater than 10,000 ohms?

Yes	GO to J21 .
No	REPAIR the circuit in question. TEST the system for normal operation.

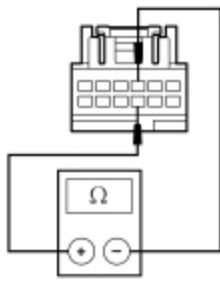
J21 CHECK VIDEO SIGNAL CIRCUITS FOR AN OPEN

- Connect a fused jumper wire between the camera switching module [C1664](#) Pin 2, circuit VMP19 (WH/GN), harness side and the camera switching module [C1664](#) Pin 3, circuit RMP19 (BN/VT), harness side.



• N0132571

- Measure the resistance between the APIM [C2383](#) Pin 14, circuit VMP19 (WH/GN), harness side and the APIM [C2383](#) Pin 15, circuit RMP19 (BN/VT), harness side.



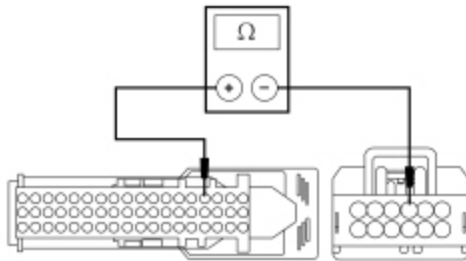
• N0082111

Is the resistance less than 5 ohms?

Yes	REMOVE the jumper wire. GO to J23 .
No	REMOVE the jumper wire. GO to J22 .

J22 CHECK VIDEO (-) SIGNAL CIRCUIT FOR AN OPEN

- Measure the resistance between the camera switching module [C1664](#) Pin 3, circuit RMP19 (BN/VT), harness side and the APIM [C2383](#) Pin 15, circuit RMP19 (BN/VT), harness side.



• N0149437

Is the resistance less than 5 ohms?

Yes	REPAIR circuit VMP19 (WH/GN). TEST the system for normal operation.
No	REPAIR circuit RMP19 (BN/VT). TEST the system for normal operation.

J23 CHECK FOR CORRECT CAMERA SWITCHING MODULE OPERATION

- Connect all disconnected connectors.
- Disconnect the camera switching module connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the camera switching module connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new camera switching module. REFER to Camera Switching Module in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test K: BCM DTC B1331:02 or B1331:08

Normal Operation

When reverse gear is selected, the BCM sends the reverse gear message to the auto-dimming interior mirror through a LIN circuit. The auto-dimming interior mirror enables the rear video camera display based on this message.

DTC Fault Trigger Conditions

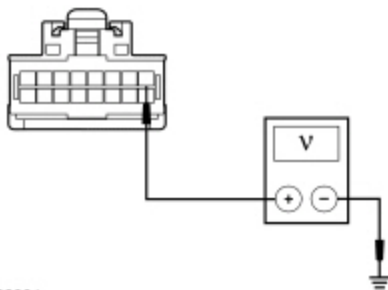
Root DTC Description	Failure Types	Fault Trigger Conditions
B1331:02	Compass/Mirror Module: General Signal Failure	Sets in continuous memory in the BCM when invalid LIN data is received from the auto-dimming interior mirror.
B1331:08	Compass/Mirror Module: Bus Signal/Message Failure	Sets in continuous memory and on-demand in the BCM when expected LIN messages are not received from the auto-dimming interior mirror.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Auto-dimming interior mirror
- BCM

PINPOINT TEST K : BCM DTC B1331:02 OR B1331:08

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
K1 CHECK THE AUTO-DIMMING INTERIOR MIRROR VOLTAGE SUPPLY
• Ignition OFF. • Disconnect: Auto-Dimming Interior Mirror C9039. • Ignition ON. • Measure the voltage between the auto-dimming interior mirror C9039 Pin 1, circuit CBP32 (GN/VT), harness side and ground.



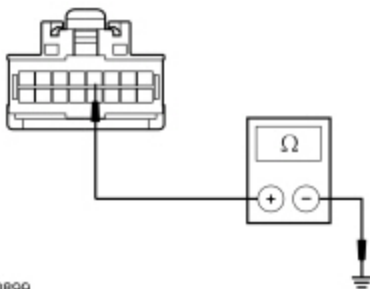
• N0068991

Is the voltage greater than 10 volts?

Yes	GO to K2 .
No	VERIFY the BCM fuse 32 (15A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. TEST the system for normal operation.

K2 CHECK THE AUTO-DIMMING INTERIOR MIRROR GROUND CIRCUIT

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Measure the resistance between the auto-dimming interior mirror [C9039](#) Pin 4, circuit GD133 (BK), harness side and ground.



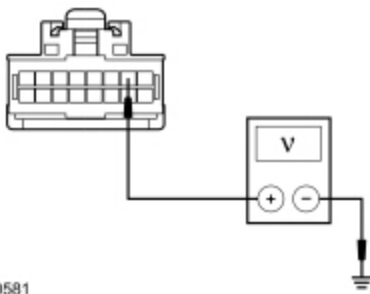
• N0120899

Is the resistance less than 5 ohms?

Yes	GO to K3 .
No	REPAIR the circuit. CONNECT the negative battery cable.

K3 CHECK THE LIN CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: BCM [C2280B](#) .
- Ignition ON.
- Measure the voltage between interior auto-dimming mirror [C9039](#) Pin 2, circuit VDN01 (BU/WH), harness side and ground.



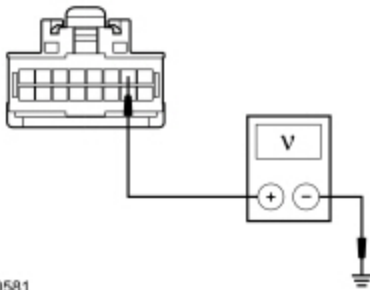
N0160581

Is any voltage present?

Yes	If the vehicle is equipped with a 110 volt AC power outlet, GO to K4 . If the vehicle is not equipped with a 110 volt AC power outlet, REPAIR the circuit.
No	GO to K5 .

K4 CHECK THE LIN CIRCUIT FOR A SHORT TO VOLTAGE WITH THE DC /AC INVERTER DISCONNECTED

- Ignition OFF.
- Disconnect: DC /AC Inverter [C2293A](#).
- Ignition ON.
- Measure the voltage between interior auto-dimming mirror [C9039](#) Pin 2, circuit VDN01 (BU/WH), harness side and ground.



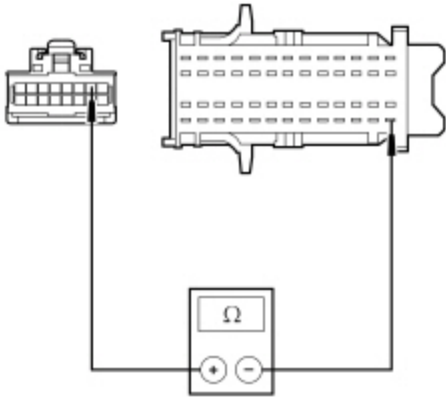
N0160581

Is any voltage present?

Yes	REPAIR the circuit.
No	GO to K8 .

K5 CHECK THE LIN CIRCUIT FOR AN OPEN

- Ignition OFF.
- Measure the resistance between interior auto-dimming mirror [C9039](#) Pin 2, circuit VDN01 (BU/WH), harness side and BCM [C2280B](#) Pin 52, circuit VDN01 (BU/WH), harness side.



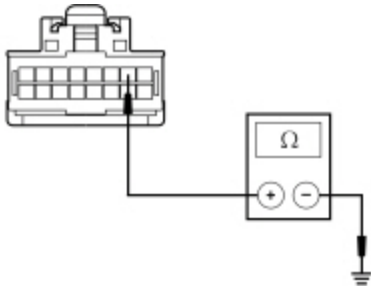
N0160584

Is the resistance less than 3 ohms?

Yes	GO to K6 .
No	REPAIR the circuit.

K6 CHECK THE LIN CIRCUIT FOR A SHORT TO GROUND

- Measure the resistance between interior auto-dimming mirror [C9039](#) Pin 2, circuit VDN01 (BU/WH), harness side and ground.



N0160582

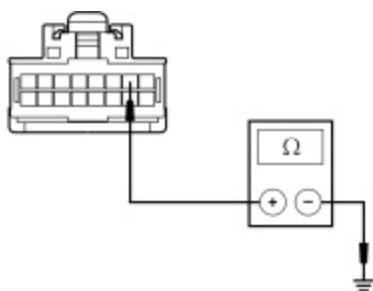
Is the resistance less than 10,000 ohms?

Yes	If the vehicle is equipped with a 110 volt AC power outlet, GO to K7 . If the vehicle is not equipped with a 110 volt AC power outlet, REPAIR the circuit.
No	INSTALL a new interior auto-dimming mirror. REFER to Section 501-09. If the concern is still present after mirror replacement, GO to K9 .

K7 CHECK THE LIN CIRCUIT FOR A SHORT TO GROUND WITH THE DC /AC INVERTER DISCONNECTED

- Ignition OFF.

- Disconnect: ~~DC~~ /~~AC~~ Inverter [C2293A](#).
- Measure the resistance between interior auto-dimming mirror [C9039](#) Pin 2, circuit VDN01 (BU/WH), harness side and ground.



N0160582

Is the resistance less than 10,000 ohms?

Yes	REPAIR the circuit.
No	GO to K8 .

K8 CHECK FOR CORRECT DC /AC INVERTER OPERATION

- Ignition OFF.
- Disconnect and inspect all ~~DC~~ /~~AC~~ Inverter connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed out pins (Install new pins as necessary)
- Reconnect the ~~DC~~ /~~AC~~ inverter connectors. Make sure they seat and latch correctly.
- Operate the system and determine If the concern is still present

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new DC / AC inverter. REFER to Section 414-05.
No	The system is operating correctly at this time. The concern may have been caused by an loose or corroded connector.

K9 CHECK FOR CORRECT BCM OPERATION

- Ignition OFF.
- Disconnect and inspect all ~~BCM~~ connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed out pins (Install new pins as necessary)

- Reconnect the BCM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present

Is the concern still present?

Yes	CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. .
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